DATA STRUCTURES AND ALGORITHMS

LECTURE 13

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In Lecture 12...

- Binary Tree
- Binary Search Tree

Today

- Binary Tree
- AVL Tree

Binary Tree Traversal

- A node of a (binary) tree is visited when the program control arrives at the node, usually with the purpose of performing some operation on the node (printing it, checking the value from the node, etc.).
- *Traversing* a (binary) tree means visiting all of its nodes.
- For a binary tree there are 4 possible traversals:
  - Preorder
  - Inorder
  - Postorder
  - Level order (breadth first) - the same as in case of a (non-binary) tree

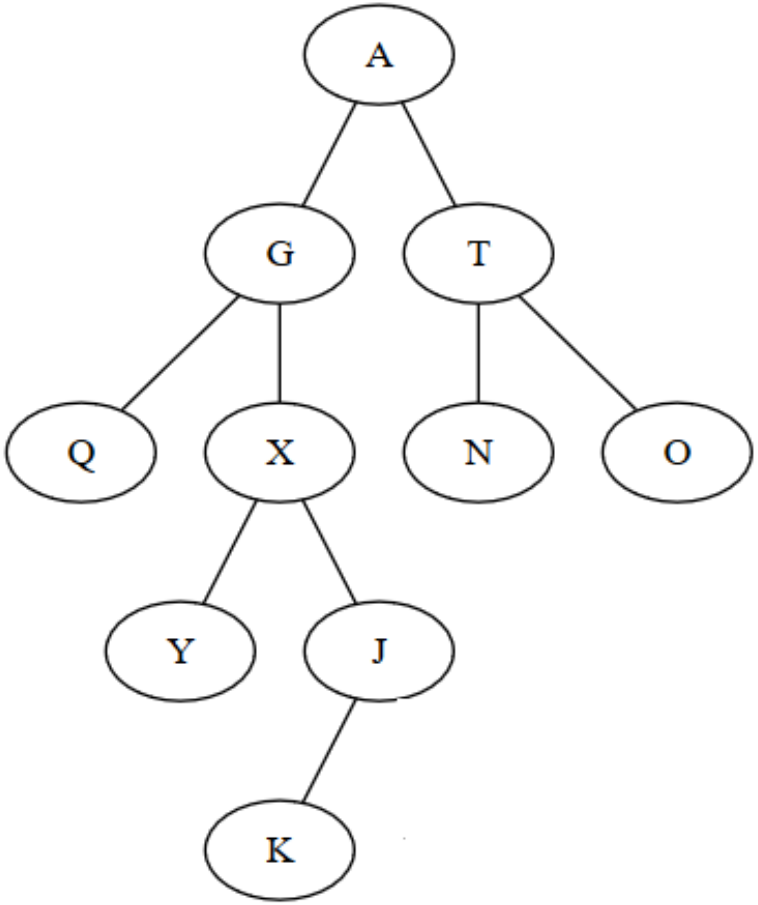
Binary tree representation

- In the following, for the implementation of the traversal algorithms, we are going to use the following representation for a binary tree:

BTNode:
  info: TElem
  left: ↑ BTNode
  right: ↑ BTNode

BinaryTree:
  root: ↑ BTNode

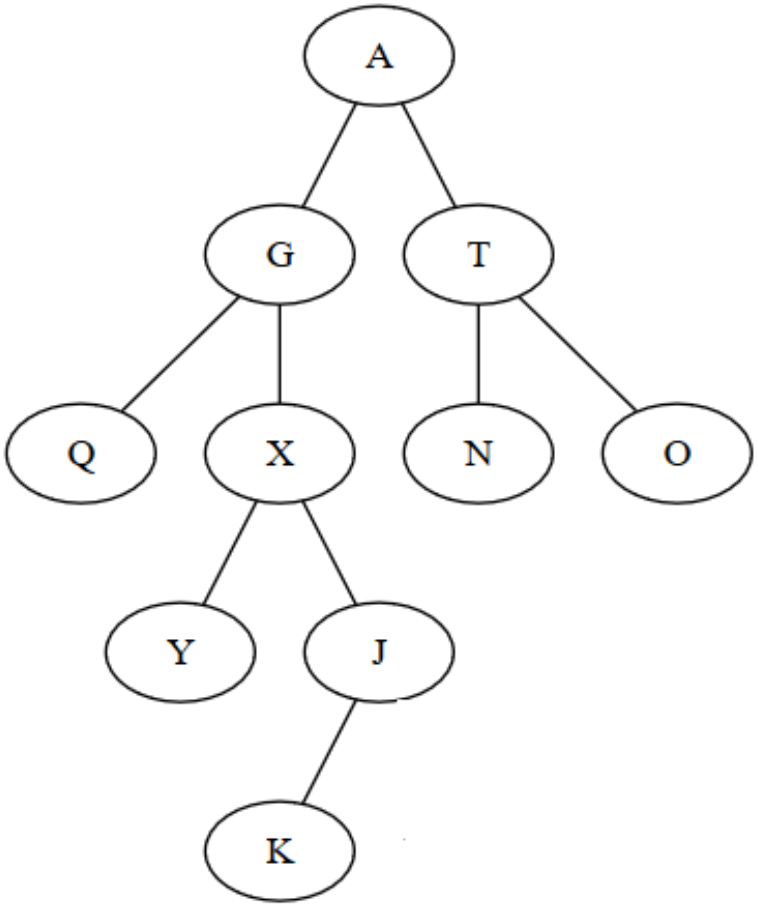
Preorder traversal example



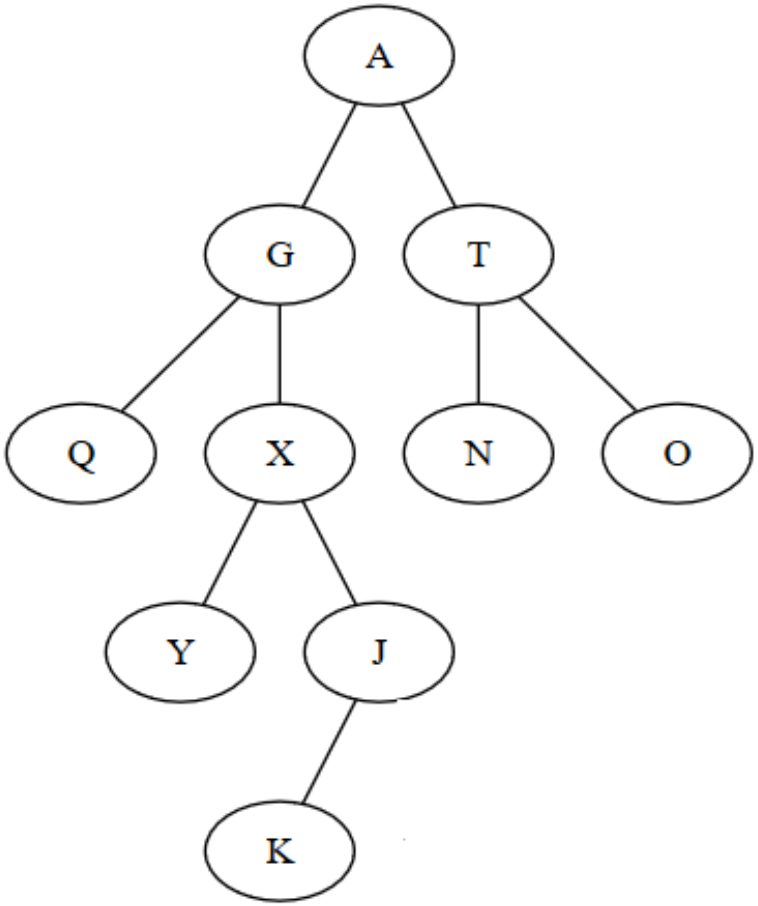
Postorder traversal

- In case of *postorder* traversal:
  - Traverse the left subtree - if exists
  - Traverse the right subtree - if exists
  - Visit the *root* of the tree
- When traversing the subtrees (left or right) the same postorder traversal is applied (so, from the left subtree we traverse the left subtree, then traverse the right subtree and then visit the root).

Postorder traversal example



Postorder traversal example



- Postorder traversal: Q, Y, K, J, X, G, N, O, T, A

Postorder traversal - recursive implementation

- The simplest implementation for postorder traversal is with a recursive algorithm.

```
subalgorithm postorder_recursive(node) is:
//pre: node is a ↑ BTNode
    if node ≠ NIL then
        postorder_recursive([node].left)
        postorder_recursive([node].right)
        @visit [node].info
    end-if
end-subalgorithm
```

- We need again a wrapper subalgorithm to perform the first call to *postorder_recursive* with the root of the tree as parameter.
- The traversal takes $\Theta(n)$ time for a tree with n nodes.

Postorder traversal - non-recursive implementation

- We can implement the postorder traversal without recursion, but it is slightly more complicated than preorder and inorder traversals.
- We can have an implementation that uses two stacks and there is also an implementation that uses one stack.

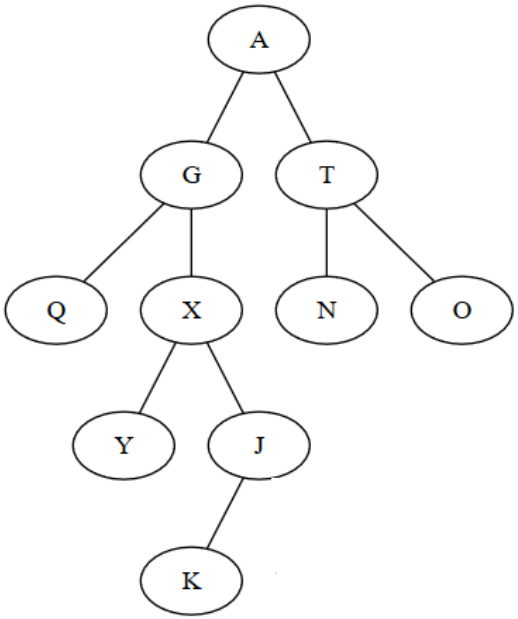
Postorder traversal with two stacks

- The main idea of postorder traversal with two stacks is to build the reverse of postorder traversal in one stack. If we have this, popping the elements from the stack until it becomes empty will give us postorder traversal.
- Building the reverse of postorder traversal is similar to building preorder traversal, except that we need to traverse the right subtree first (not the left one). The other stack will be used for this.
- The algorithm is similar to *preorder* traversal, with two modifications:
  - When a node is removed from the stack, it is added to the second stack (instead of being visited)
  - For a node taken from the stack we first push the left child and then the right child to the stack.

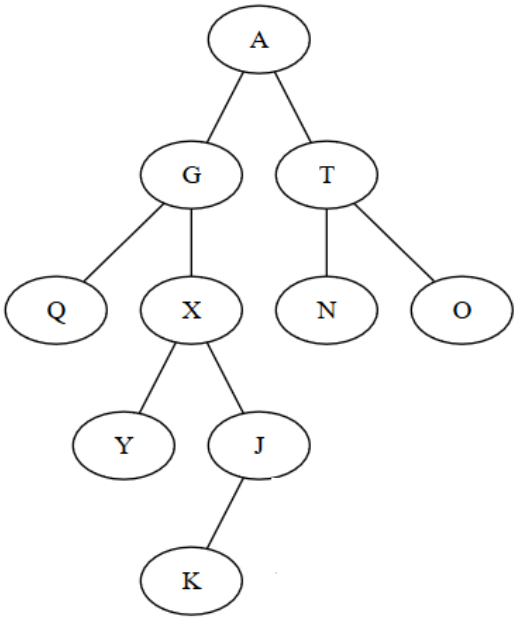
Postorder traversal with one stack

- We start with an empty stack and a current node set to the root of the tree
- While the current node is not NIL, push to the stack the right child of the current node (if exists) and the current node and then set the current node to its left child.
- While the stack is not empty
  - Pop a node from the stack (call it current node)
  - If the current node has a right child, the stack is not empty and contains the right child on top of it, pop the right child, push the current node, and set current node to the right child.
  - Otherwise, visit the current node and set it to NIL
  - While the current node is not NIL, push to the stack the right child of the current node (if exists) and the current node and then set the current node to its left child.

Postorder traversal - non-recursive implementation example



Postorder traversal - non-recursive implementation example



- Node: A (Stack:)
- Node: NIL (Stack: T A X G Q)
- Visit Q, Node NIL (Stack: T A X G)
- Node: X (Stack: T A G)
- Node: NIL (Stack: T A G J X Y)
- Visit Y, Node: NIL (Stack: T A G J X)
- Node: J (Stack: T A G X)
- Node: NIL (Stack: T A G X J K)
- Visit K, Node: NIL (Stack: T A G X J)
- Visit J, Node: NIL (Stack: T A G X)
- Visit X, Node: NIL (Stack: T A G)
- Visit G, Node: NIL (Stack: T A)
- Node: T (Stack: A)
- Node: NIL (Stack: A O T N)
- ...

Postorder traversal - non-recursive implementation

```
subalgorithm postorder(tree) is:
//pre: tree is a BinaryTree
    s: Stack //s is an auxiliary stack
    node ← tree.root
    while node ≠ NIL execute
        if [node].right ≠ NIL then
            push(s, [node].right)
        end-if
        push(s, node)
        node ← [node].left
    end-while
    while not isEmpty(s) execute
        node ← pop(s)
        if [node].right ≠ NIL and (not isEmpty(s)) and [node].right = top(s) th
            pop(s)
            push(s, node)
            node ← [node].right
//continued on the next slide
```

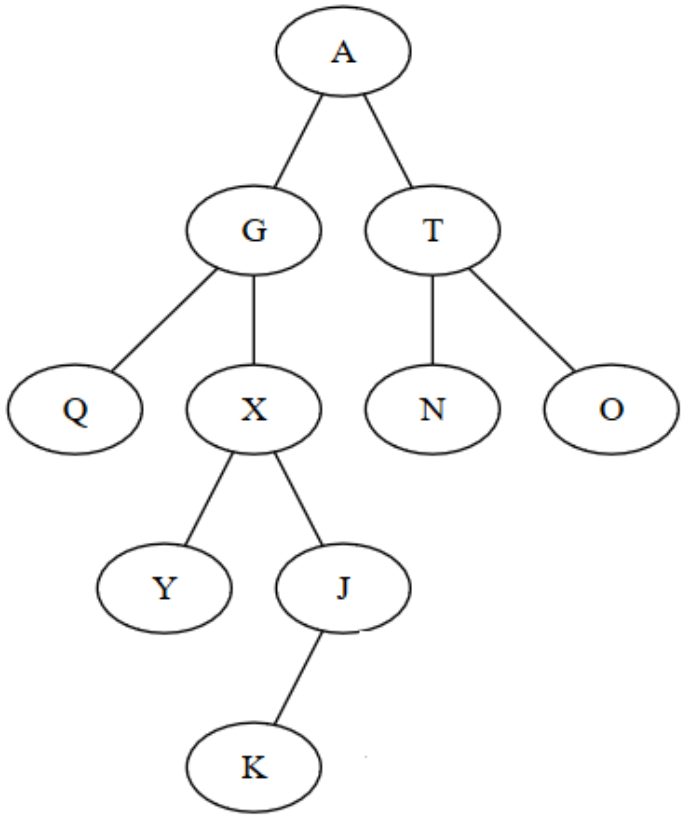
Postorder traversal - non-recursive implementation

```
        else
            @visit node
            node ← NIL
        end-if
        while node ≠ NIL execute
            if [node].right ≠ NIL then
                push(s, [node].right)
            end-if
            push(s, node)
            node ← [node].left
        end-while
    end-while
end-subalgorithm
```

- Time complexity $\Theta(n)$, extra space complexity $O(n)$

Level order traversal

- In case of level order traversal we first visit the root, then the children of the root, then the children of the children, etc.



- Level order traversal: A, G, T, Q, X, N, O, Y, J, K

Binary tree iterator

- The interface of the binary tree contains the *iterator* operation, which should return an iterator.
- This operation receives a parameter that specifies what kind of traversal we want to do with the iterator (preorder, inorder, postorder, level order)
- The traversal algorithms discussed so far, traverse all the elements of the binary tree at once, but an iterator has to do element-by-element traversal.
- For defining an iterator, we have to divide the code into the functions of an iterator: *init*, *getCurrent*, *next*, *valid*

Preorder, Inorder, Postorder

- How to remember the difference between traversals?
  - Left subtree is always traversed before the right subtree.
  - The visiting of the root is what changes:
    - PREorder - visit the root before the left and right
    - INorder - visit the root between the left and right
    - POSTorder - visit the root after the left and right

Think about it

- Assume you have a binary tree, but you do not know how it looks like, but you have the *preorder* and *inorder* traversal of the tree. Give an algorithm for building the tree based on these two traversals.

- For example:
  - Preorder: A B F G H E L M
  - Inorder: B G F H A L E M

Think about it

- Can you rebuild the tree if you have the *postorder* and the *inorder* traversal?
- Can you rebuild the tree if you have the *preorder* and the *postorder* traversal?

Huffman coding I

- We have a message (made of characters) and we want to encode it as a binary string.
- The first options is to assign to each possible character a binary code of the same length, l. When encoding the message, simply replace every character by the corresponding binary code. When decoding a binary message, split it into pieces of length l and decode every piece to a character. It is simple, but it might lead to a long code, since the length l, depends on the number of possible characters.

Huffman coding II

- A second option is to assign to each possible character a binary code of (possibly) different length, such that characters occurring frequently in the message have shorter code. Encoding is still simple, but decoding becomes more complicated, since we do not know where to split the binary string. To make decoding possible, we need the binary codes defined in such a way that none of them is the prefix of another. Huffman encoding tells us, how to define the codes, such that the total length of the encoded message is the minimum.

Huffman coding

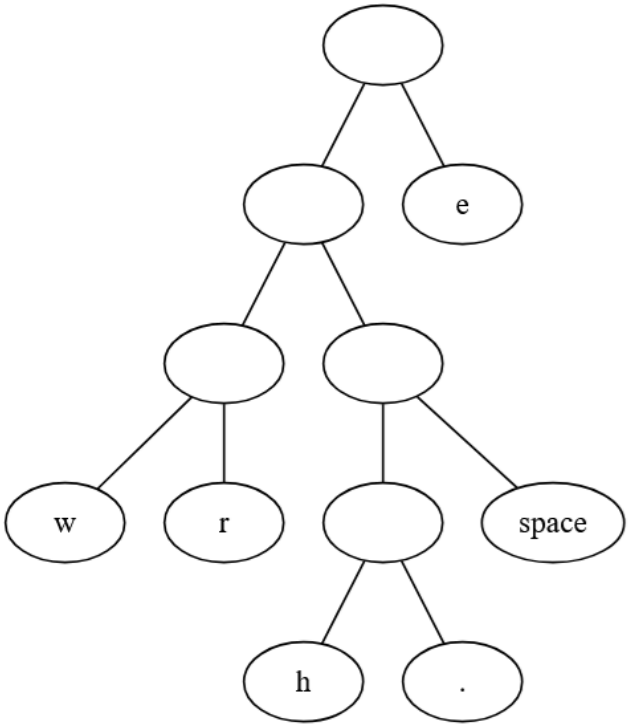
- When building the Huffman encoding, we need to have a message that we want to encode.
- First we have to compute the frequency of every character from the message, because we are going to define the codes based on the frequencies.
- Assume that we have the following message: WE WERE HERE.
- It contains the following characters and frequencies:

w	e	r	h	space	.
2	5	2	1	2	1

Huffman coding

- For defining the Huffman code a binary tree is build in the following way:
  - Start with trees containing only a root node, one for every character. Each tree has a weight, which is the frequency of the character.
  - Get the two trees with the least weight (if there is a tie, choose randomly), combine them into one tree which has as weight the sum of the two weights.
  - Repeat until we have only one tree.
- The implementation can simply be done with a Priority Queue which stores these partially built trees (and considers the weight as priority).

Huffman coding



Huffman coding

- Code for each character can be read from the tree in the following way: start from the root and go towards the corresponding leaf node. Every time we go left add the bit 0 to encoding and when we go right add bit 1.
- Code for the characters:
  - w - 000
  - r - 001
  - h - 0100
  - . - 0101
  - space - 011
  - e - 1
- We can see that the most frequent character (e), indeed has the shortest code, and the least frequent ones have the longest. Also, no code is the prefix of another.
- In order to encode a message, just replace each character with the corresponding code.

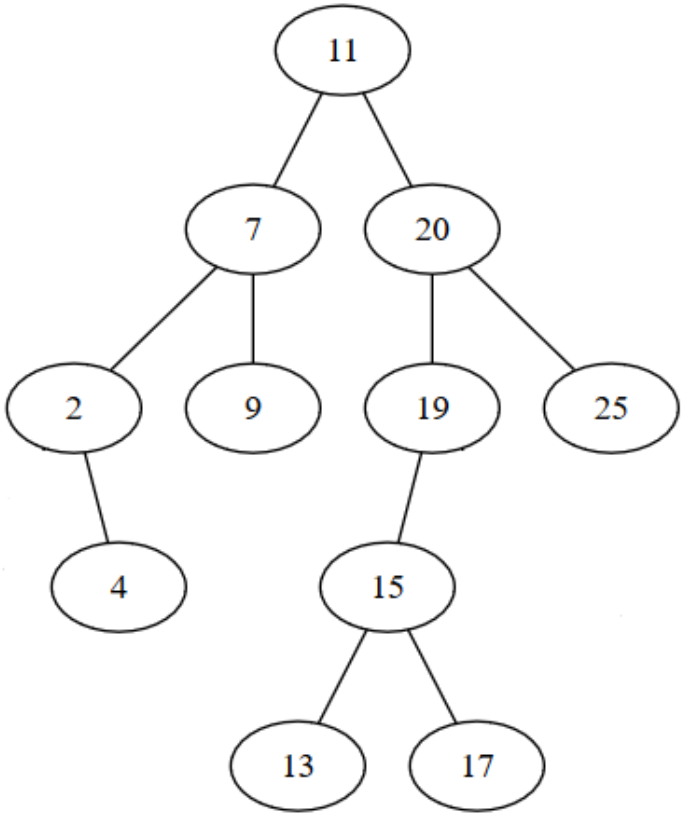
Huffman coding

- Assume we have the following code and we want to decode it: 011011000100010011001000000
- We do not know where the code of each character ends, but we can use the previously built tree to decode it.
- Start parsing the code and iterate through the tree in the following way:
  - Start from the root
  - If the current bit from the code is 0 go to the left child, otherwise go to the right child
  - If we are at a leaf node we have decoded a character and have to start over from the root
- The decoded message (quotation marks added to see the spaces at the beginning): " wewereerww"

Binary search trees - recap

- A *Binary Search Tree* is a binary tree that satisfies the following property:
  - if x is a node of the binary search tree then:
    - For every node y from the left subtree of x, the information from y is less than or equal to the information from x
    - For every node y from the right subtree of x, the information from y is greater than the information from x
- Obviously, the relation used to order the nodes can be considered in an abstract way (instead of having "$\leq$" as in the definition).

Binary Search Tree Example



- If we do an inorder traversal of a binary search tree, we will get the elements in increasing order (according to the relation used).

Balanced Binary Search Trees

- Specific operations for binary trees run in $O(h)$ time, which can be $\Theta(n)$ in worst case
- Best case is a balanced tree, where height of the tree is $\Theta(log_2 n)$

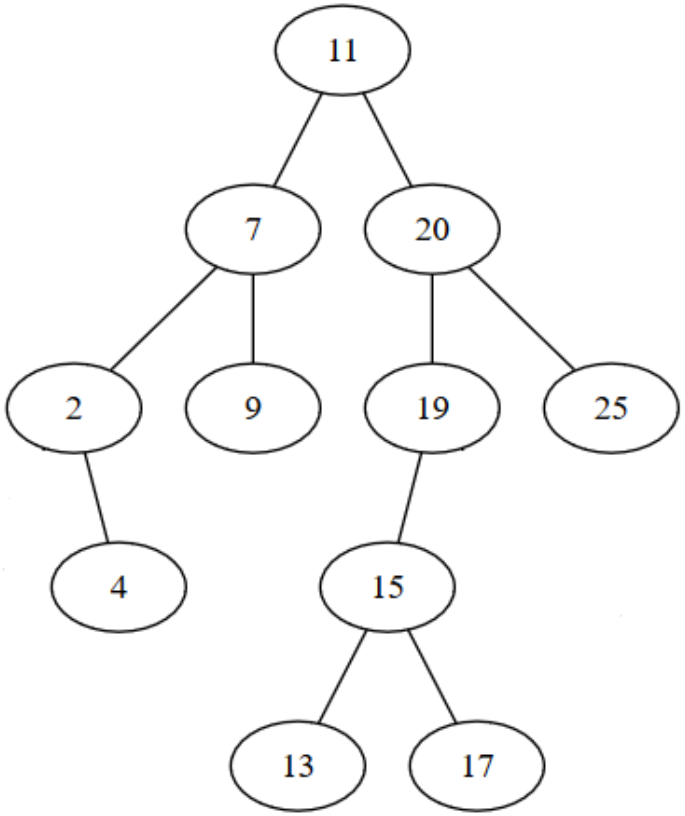
Balanced Binary Search Trees

- Specific operations for binary trees run in $O(h)$ time, which can be $\Theta(n)$ in worst case
- Best case is a balanced tree, where height of the tree is $\Theta(log_2 n)$
- To reduce the complexity of algorithms, we want to keep the tree balanced. In order to do this, we want every node to be balanced.
- When a node loses its balance, we will perform some operations (called rotations) to make it balanced again.

AVL Trees

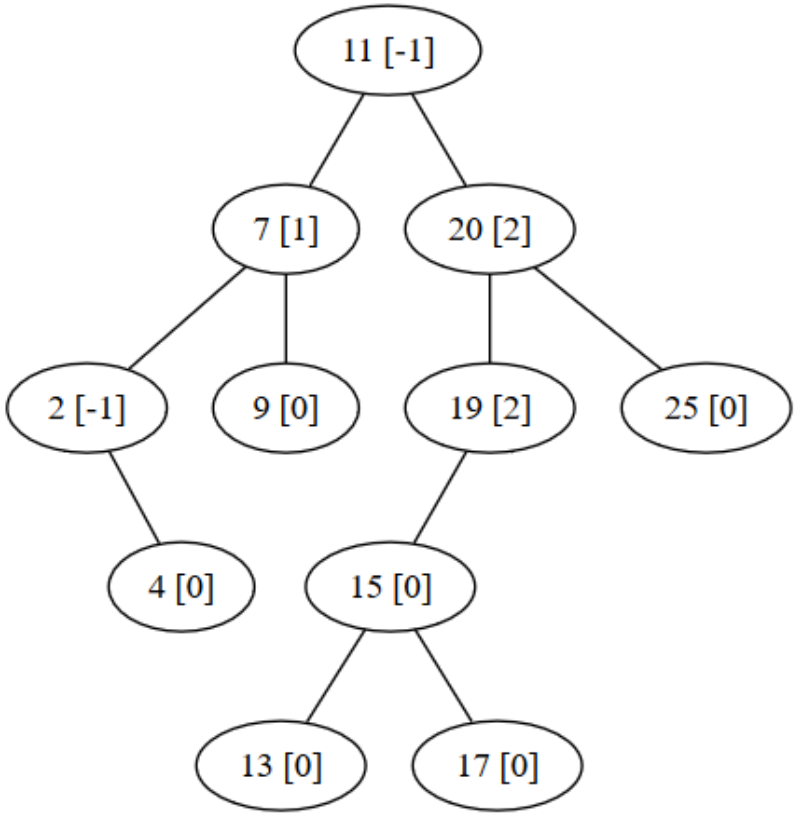
- Definition: An AVL (Adelson-Velskii Landis) tree is a binary tree which satisfies the following property (AVL tree property):
  - If x is a node of the AVL tree:
    - the difference between the height of the left and right subtree of x is 0, 1 or -1 (balancing information)
- Observations:
- Height of an empty tree is -1
- Height of a single node is 0

AVL Trees



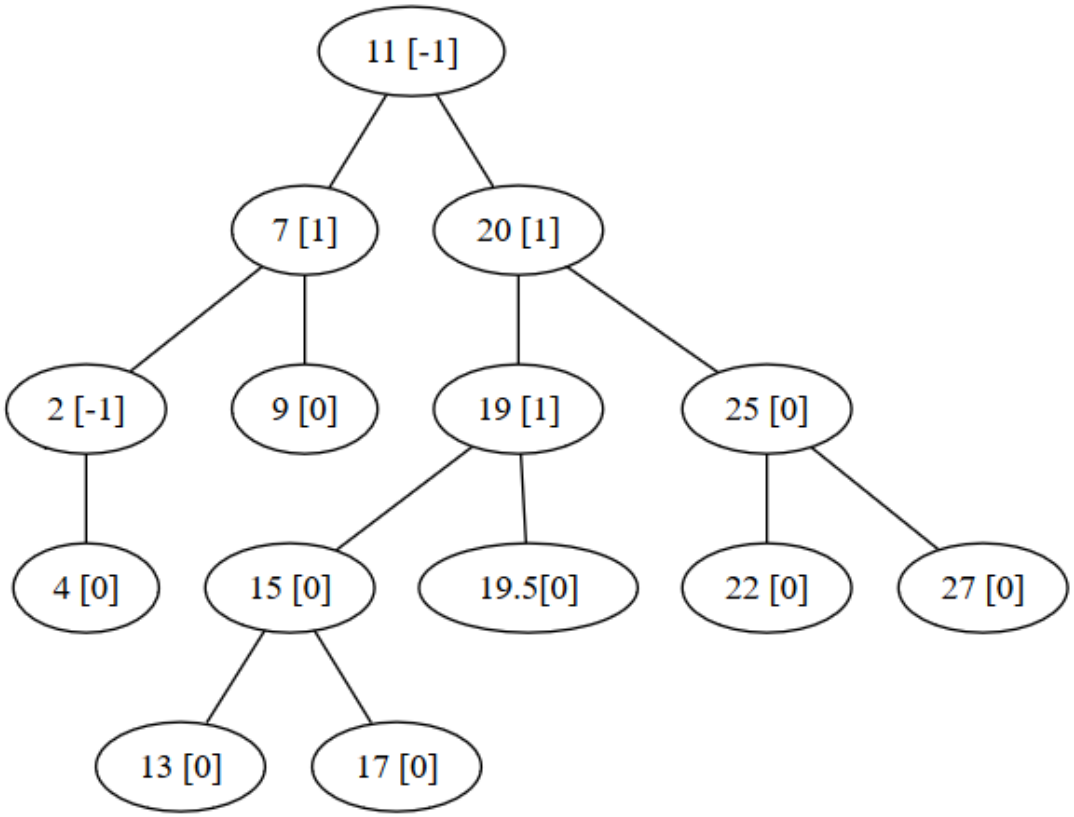
- Is this an AVL tree?

AVL Trees



- Values in square brackets show the balancing information of a node. The tree is not an AVL tree, because the balancing information for nodes 19 and 20 is 2.

AVL Trees



- This is an AVL tree.

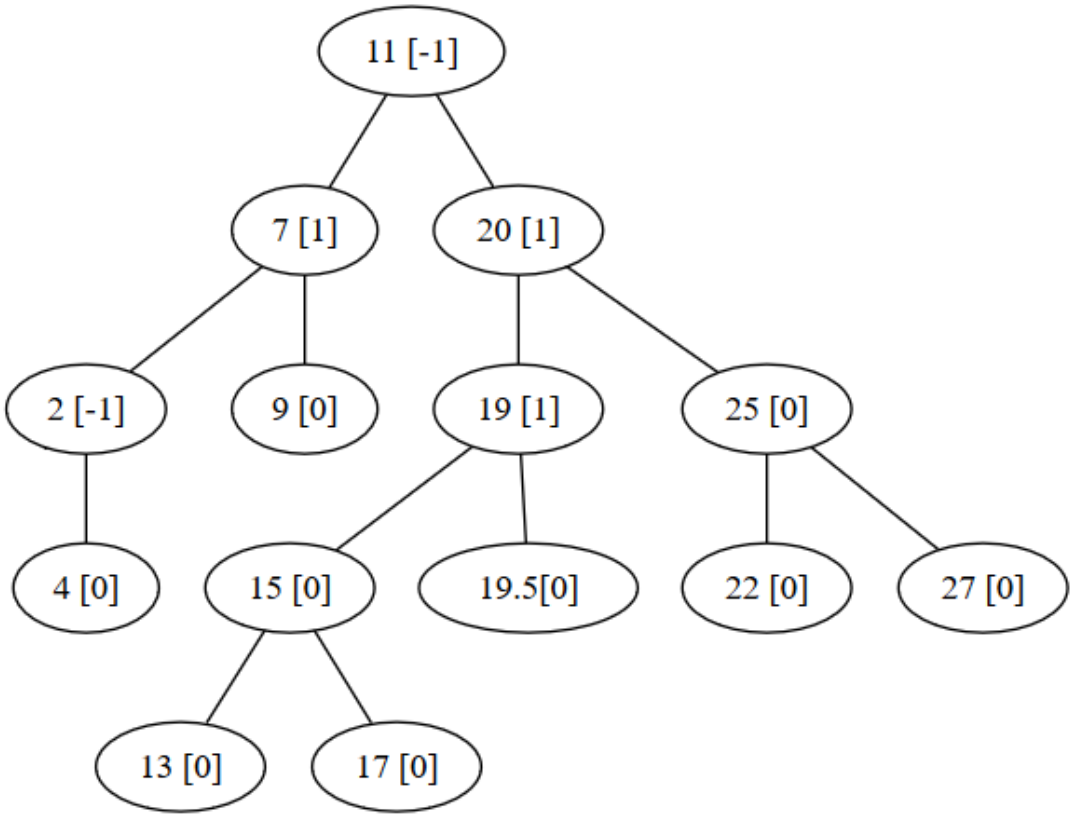
AVL Trees - rotations

- Adding or removing a node might result in a binary tree that violates the AVL tree property.
- In such cases, the property has to be restored and only after the property holds again is the operation (add or remove) considered finished.
- The AVL tree property can be restored with operations called **rotations**.

AVL Trees - rotations

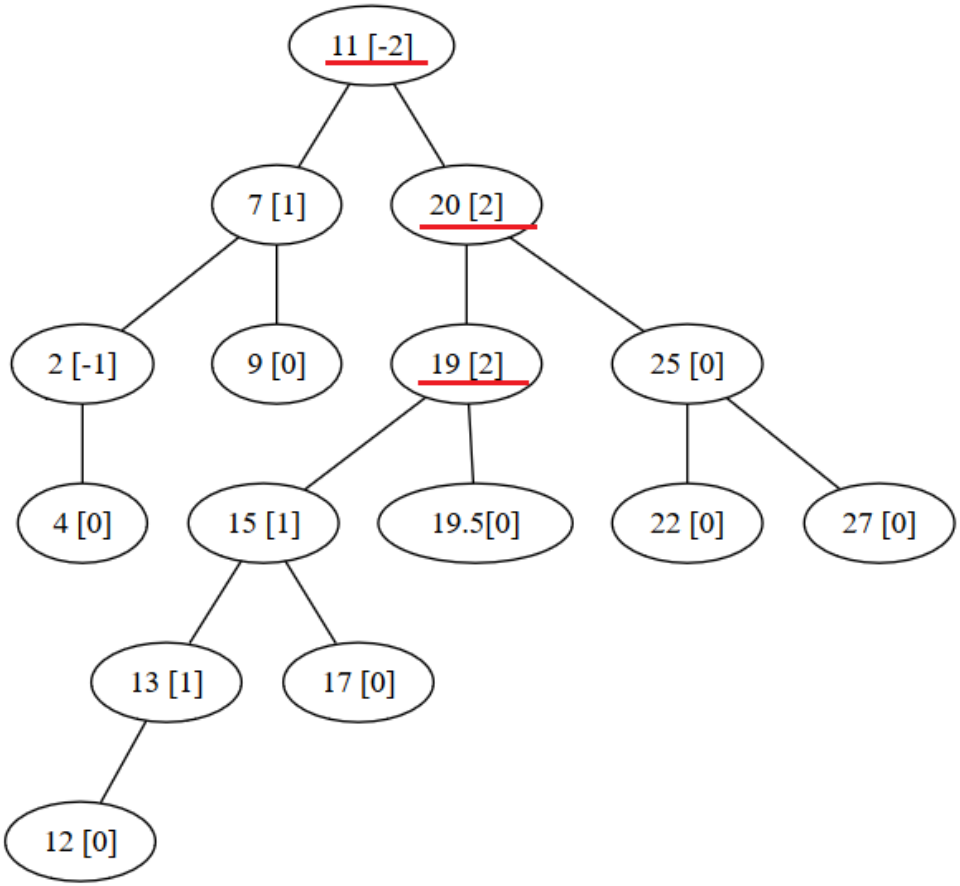
- Inserting in an AVL tree is similar to inserting in a BST: we need to find the position where the new element needs to be added and add the element there. However, this will change the balancing information of some nodes.
- After an insertion, only the nodes on the path to the modified node can change their height.
- We check the balancing information on the path from the modified node to the root (so from bottom to the top). When we find a node that does not respect the AVL tree property, we perform a suitable *rotation* to rebalance the (sub)tree.

AVL Tress - rotations



- What if we insert element 12?

AVL Trees - rotations

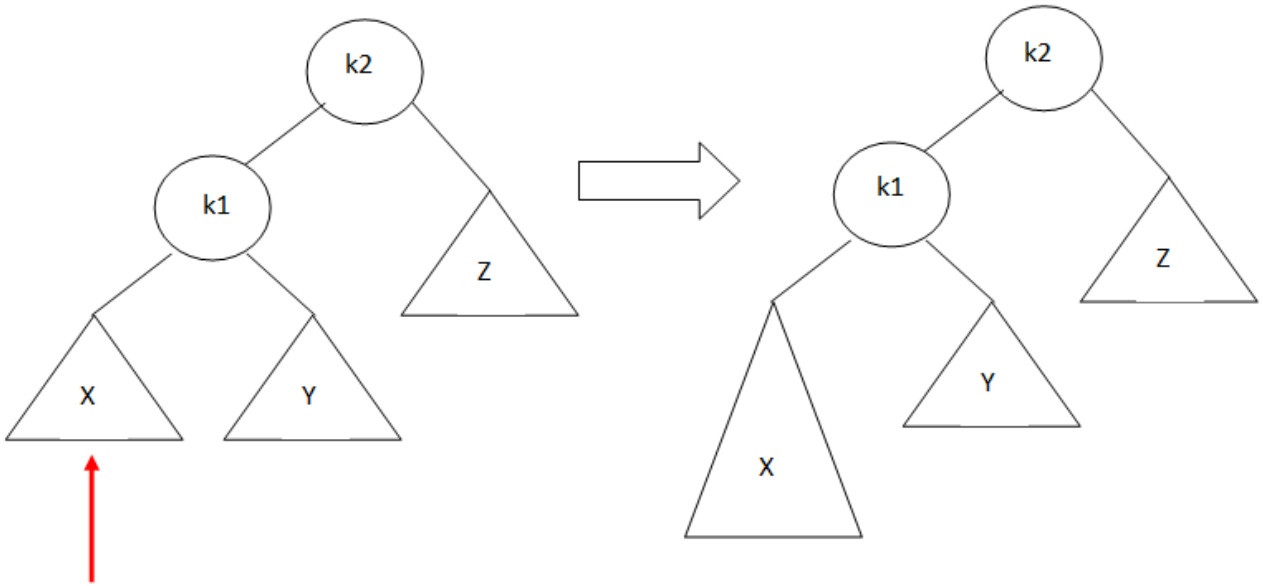


- Red lines show the unbalanced nodes. We will rebalance node 19.

AVL Trees - rotations

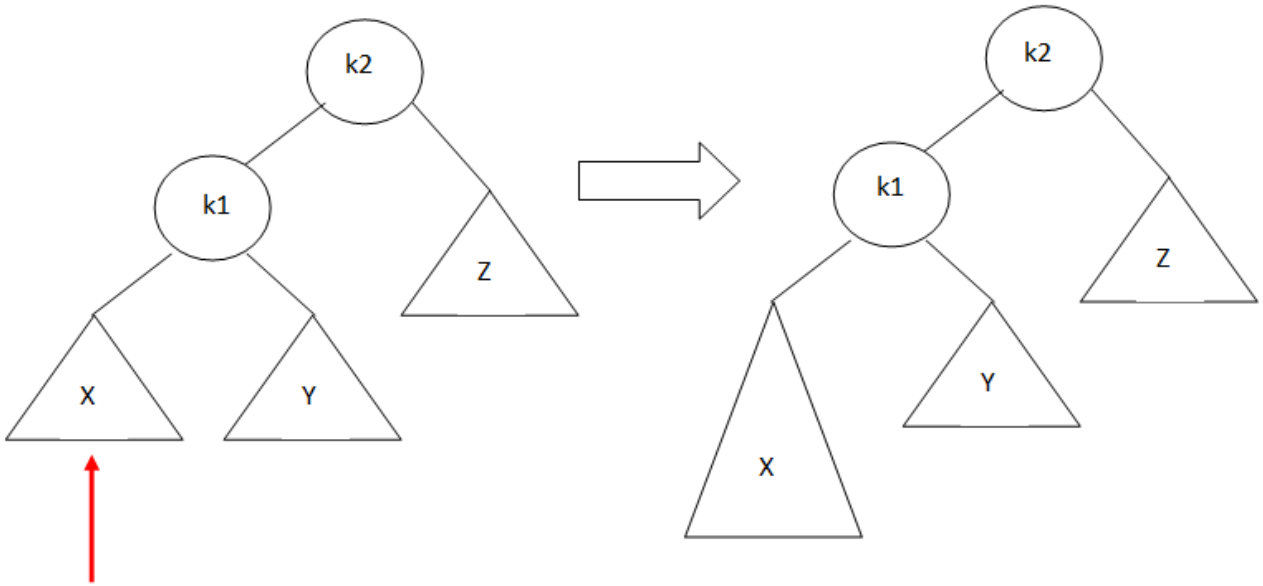
- Assume that at a given point α is the node that needs to be rebalanced. Obviously, α is not a leaf node, it has either a left or a right child, or both.
- Since α was balanced before the insertion, and is not after the insertion, we know that a new node was inserted somewhere in a subtree of α and we can identify four cases in which a violation might occur:
  - Insertion into the left subtree of the left child of α
  - Insertion into the right subtree of the left child of α
  - Insertion into the left subtree of the right child of α
  - Insertion into the right subtree of the right child of α

AVL Trees - rotations - case 1



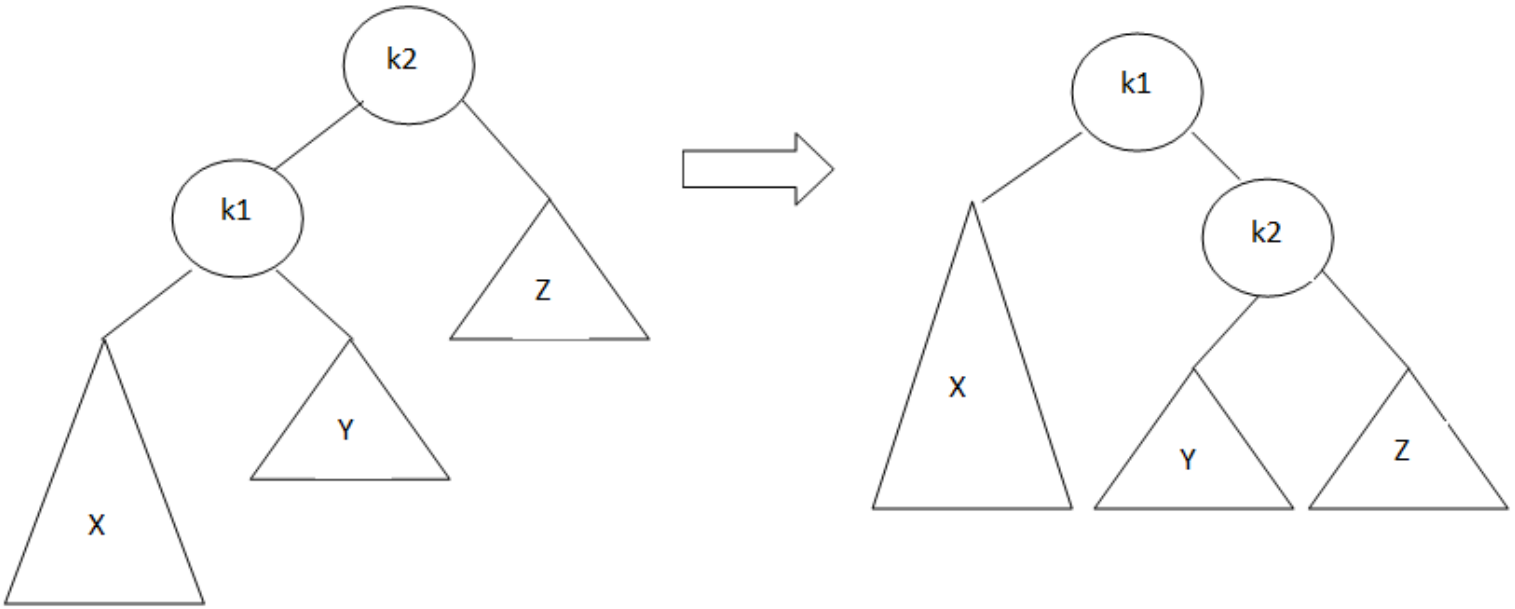
- Obs: X, Y and Z represent subtrees with the same height.

AVL Trees - rotations - case 1

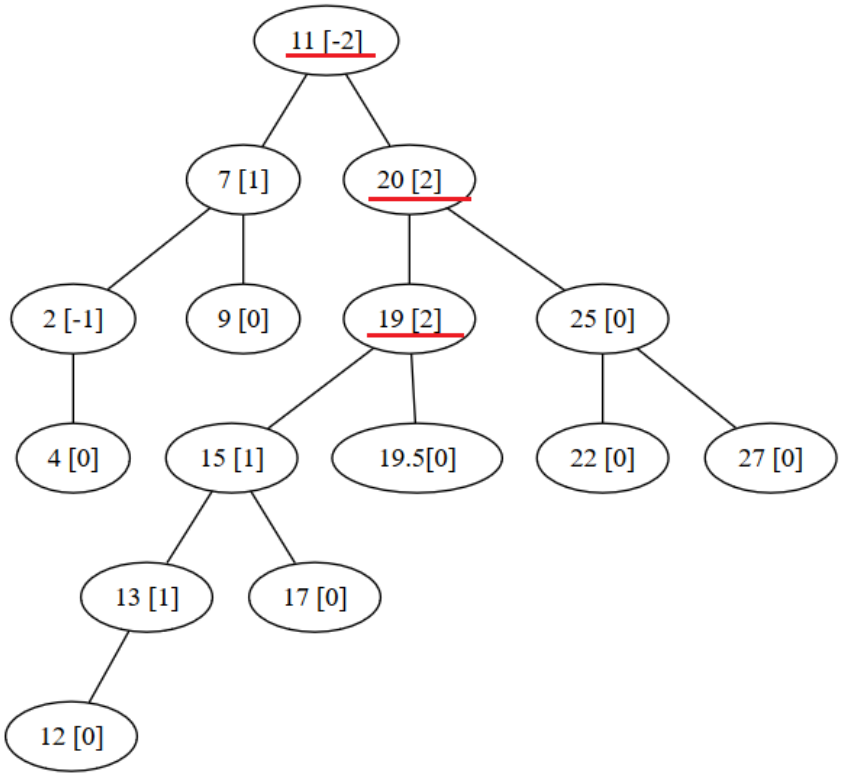


- Obs: X, Y and Z represent subtrees with the same height.
- Solution: single rotation to right

AVL Trees - rotation - Single Rotation to Right

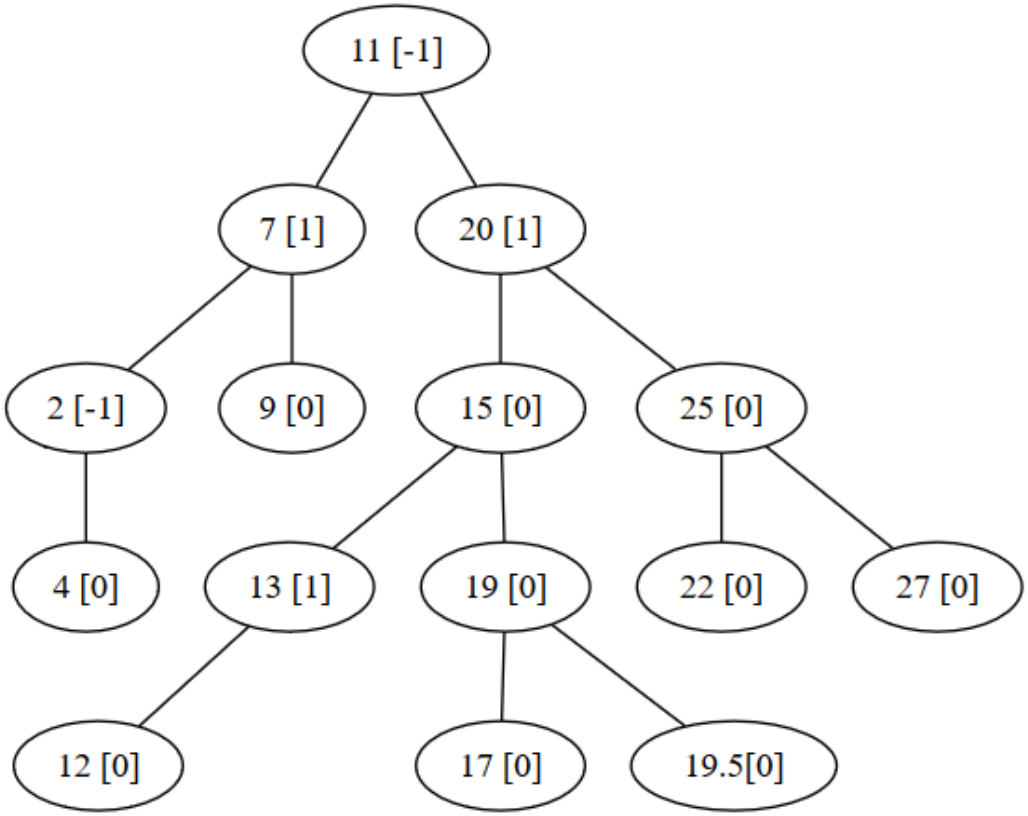


AVL Trees - rotations - case 1 example

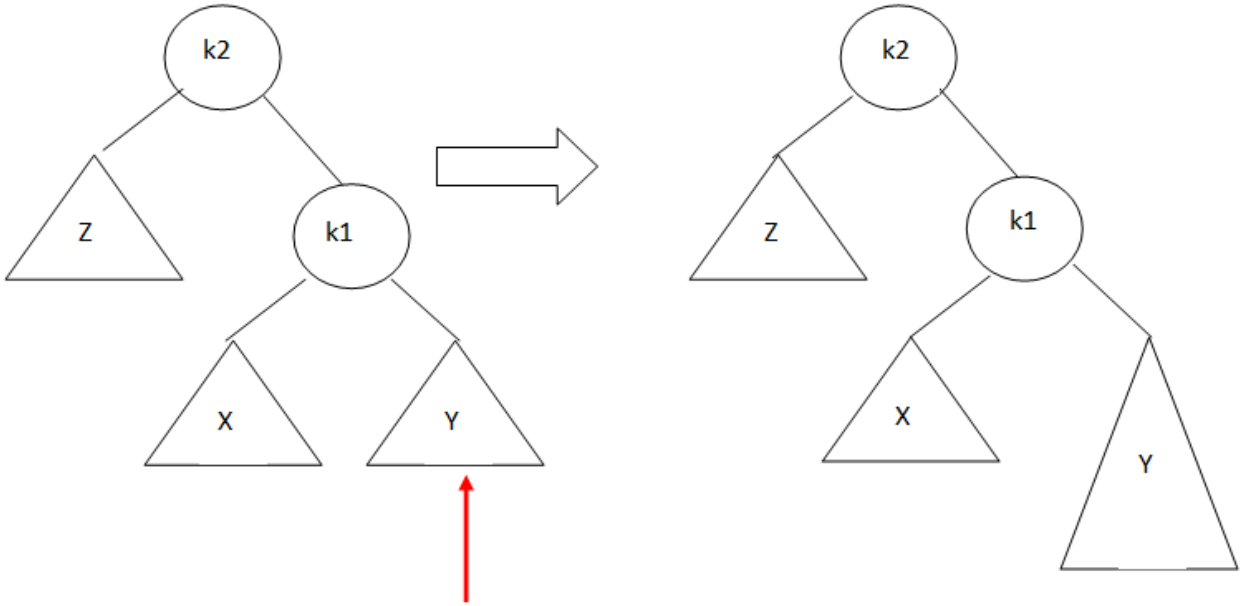


- Node 19 is imbalanced, because we inserted a new node (12) in the left subtree of the left child.
- Solution: **single rotation to right**

AVL Trees - rotation - case 1 example

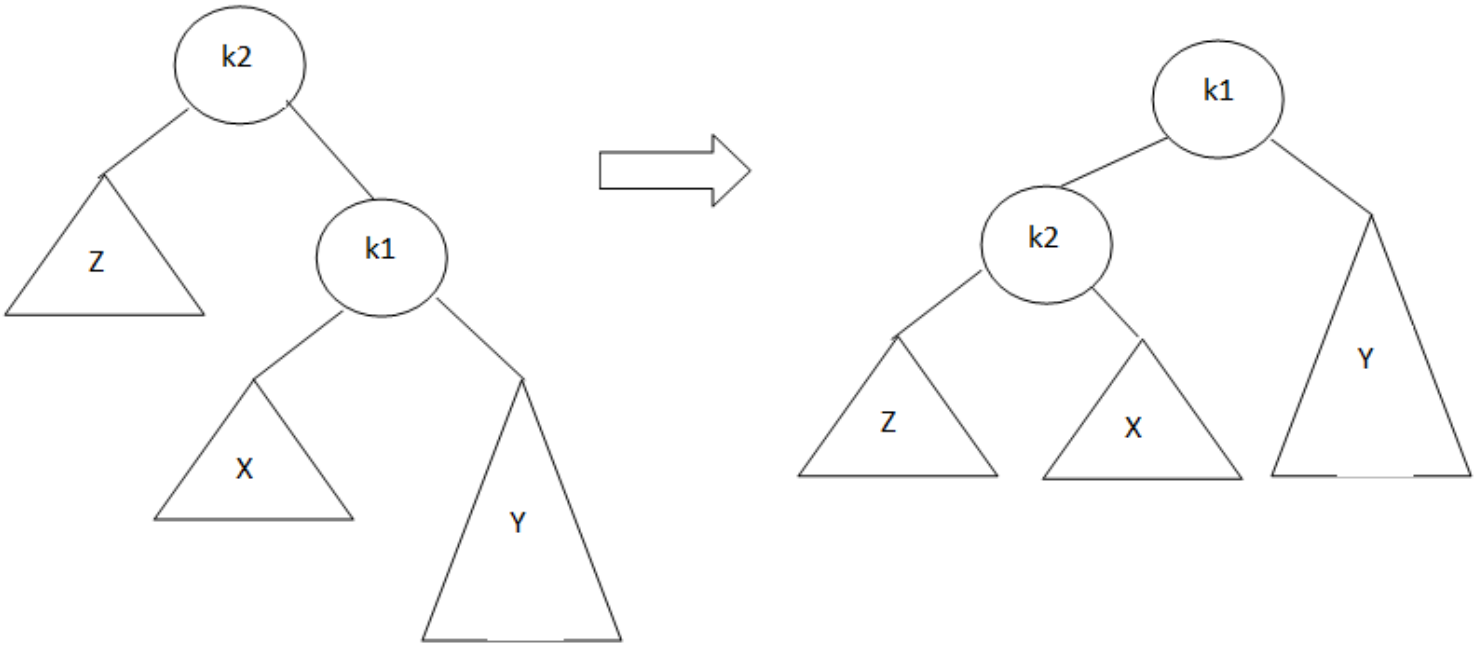


AVL Trees - rotations - case 4

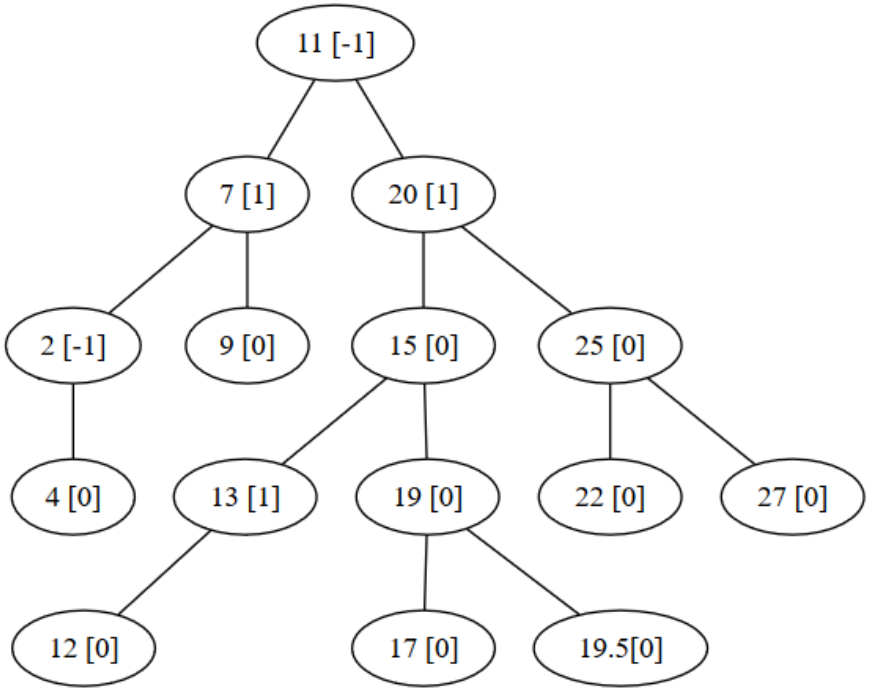


- Solution: **single rotation to left**

AVL Trees - rotation - Single Rotation to Left

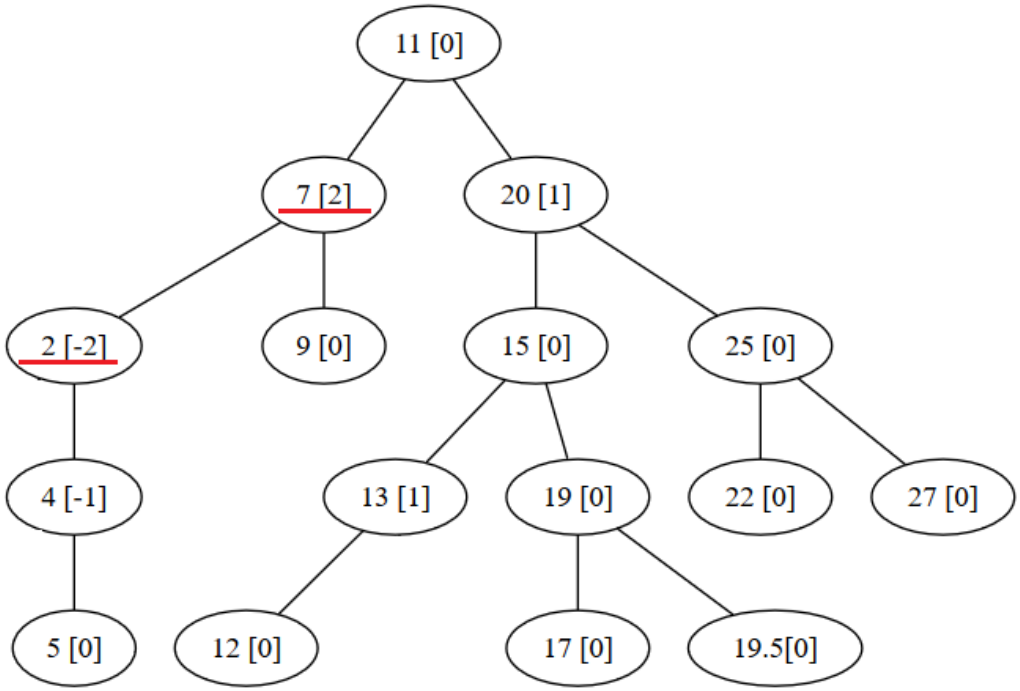


AVL Trees - rotations - case 4 example



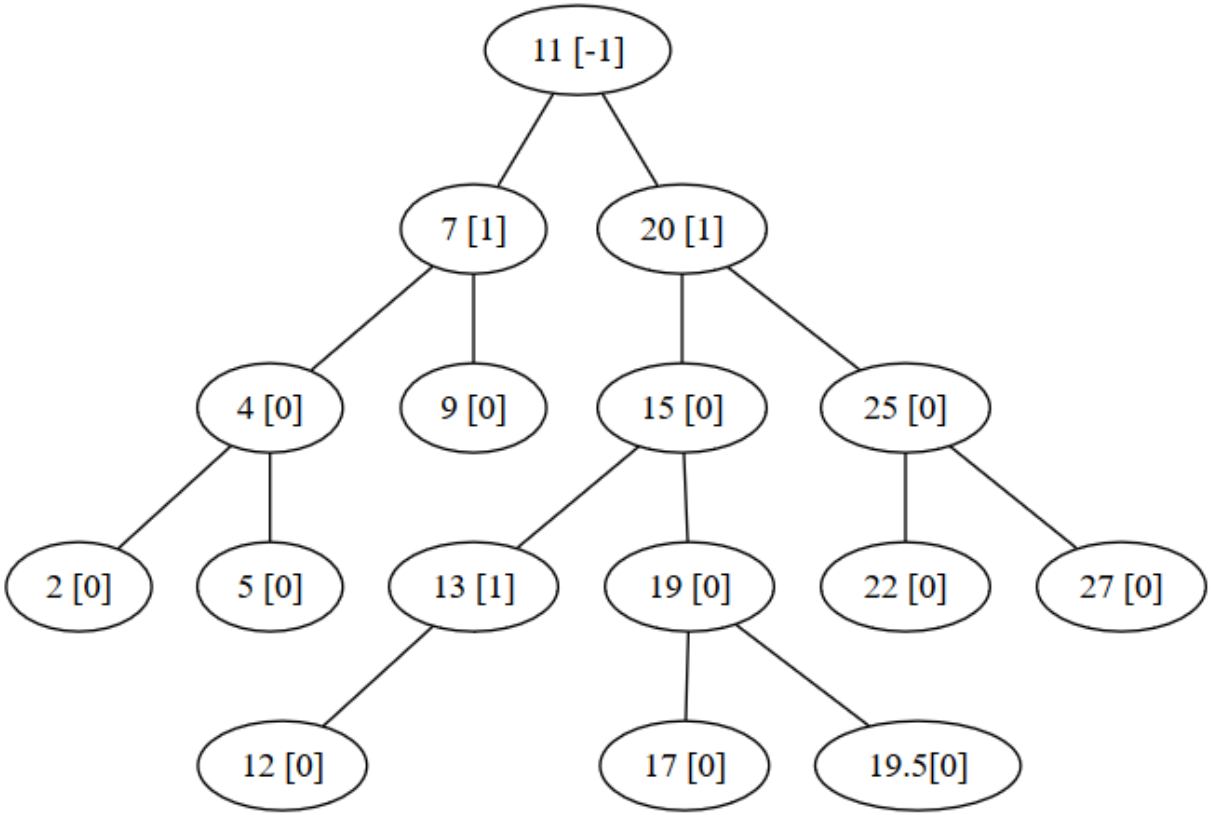
- Insert value 5

AVL Trees - rotations - case 4 example



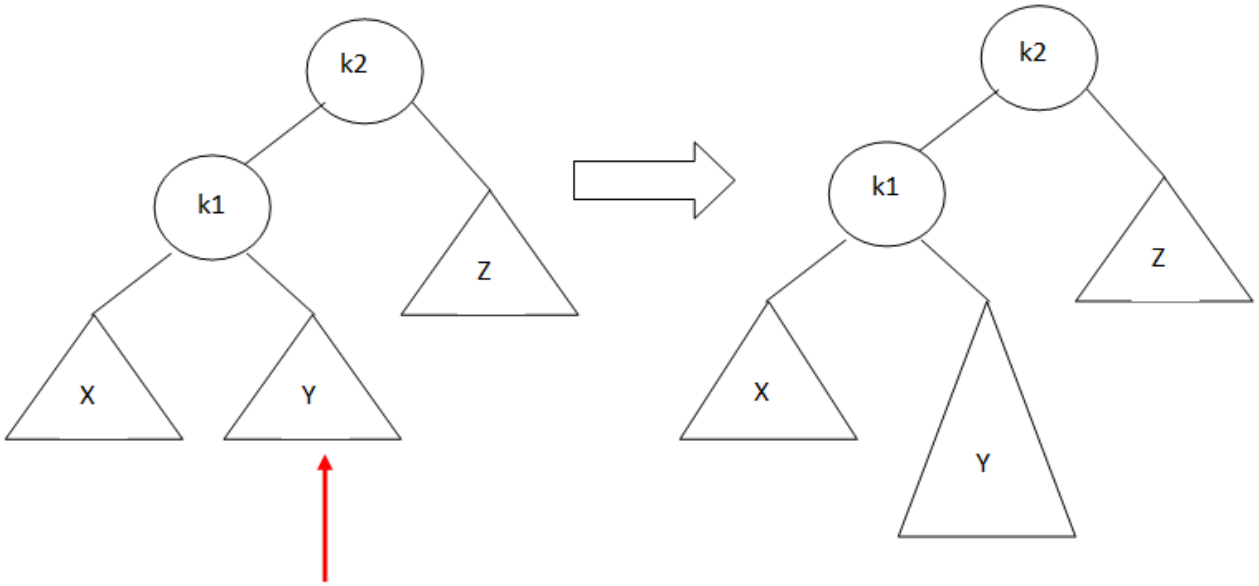
- Node 2 is imbalanced, because we inserted a new node (5) to the right subtree of the right child
- Solution: **single rotation to left**

AVL Trees - rotation - case 4 example



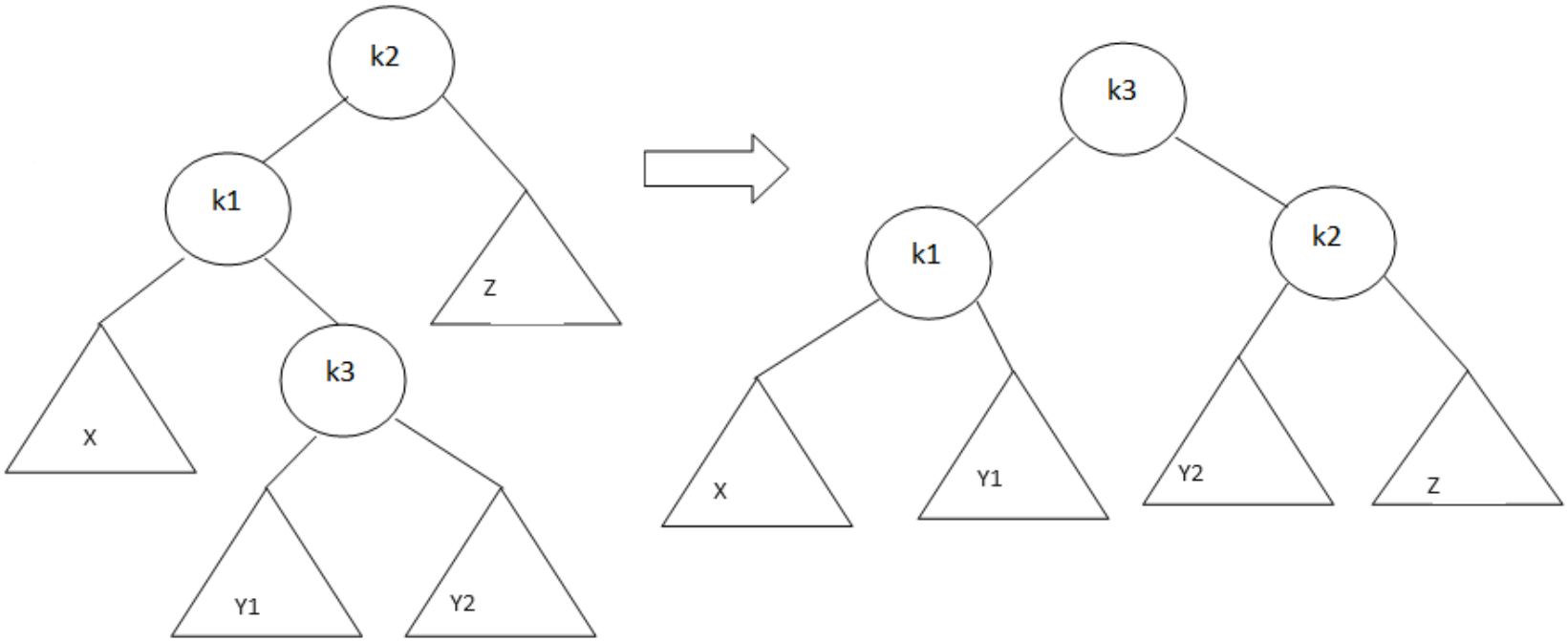
- After the rotation

AVL Trees - rotations - case 2

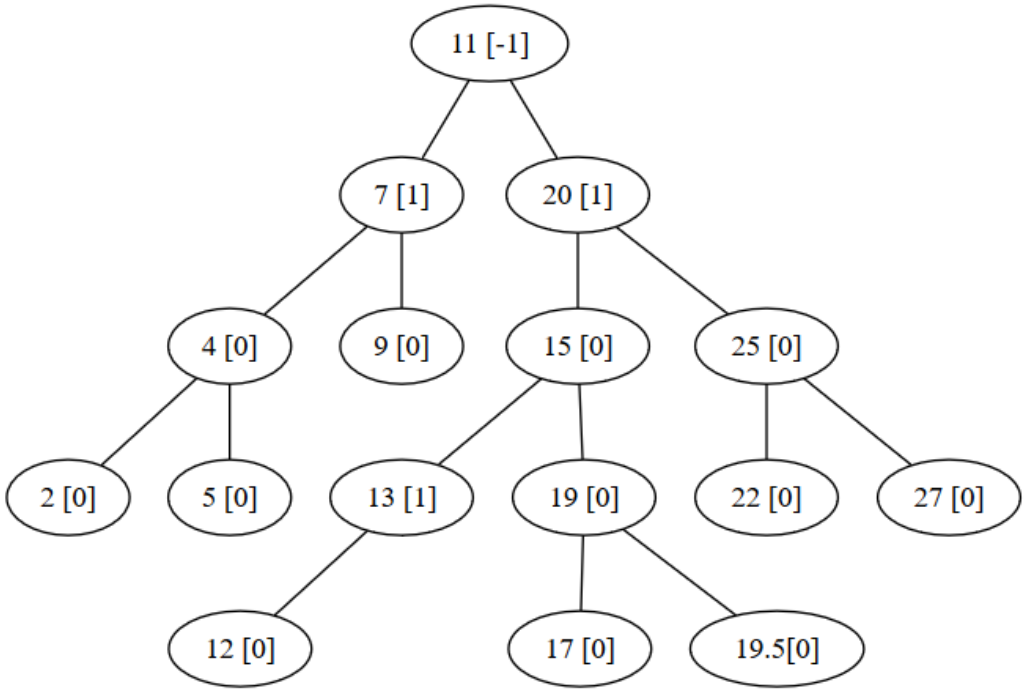


- Solution: **Double rotation to right**

AVL Trees - rotation - Double Rotation to Right

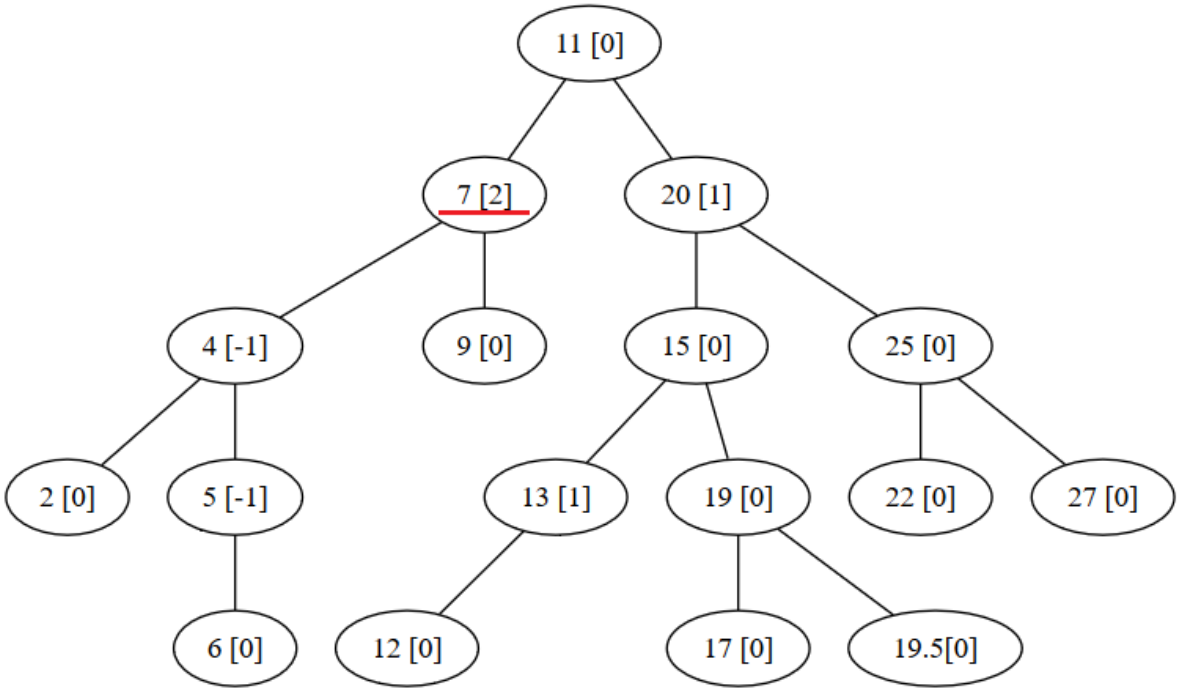


AVL Trees - rotations - case 2 example



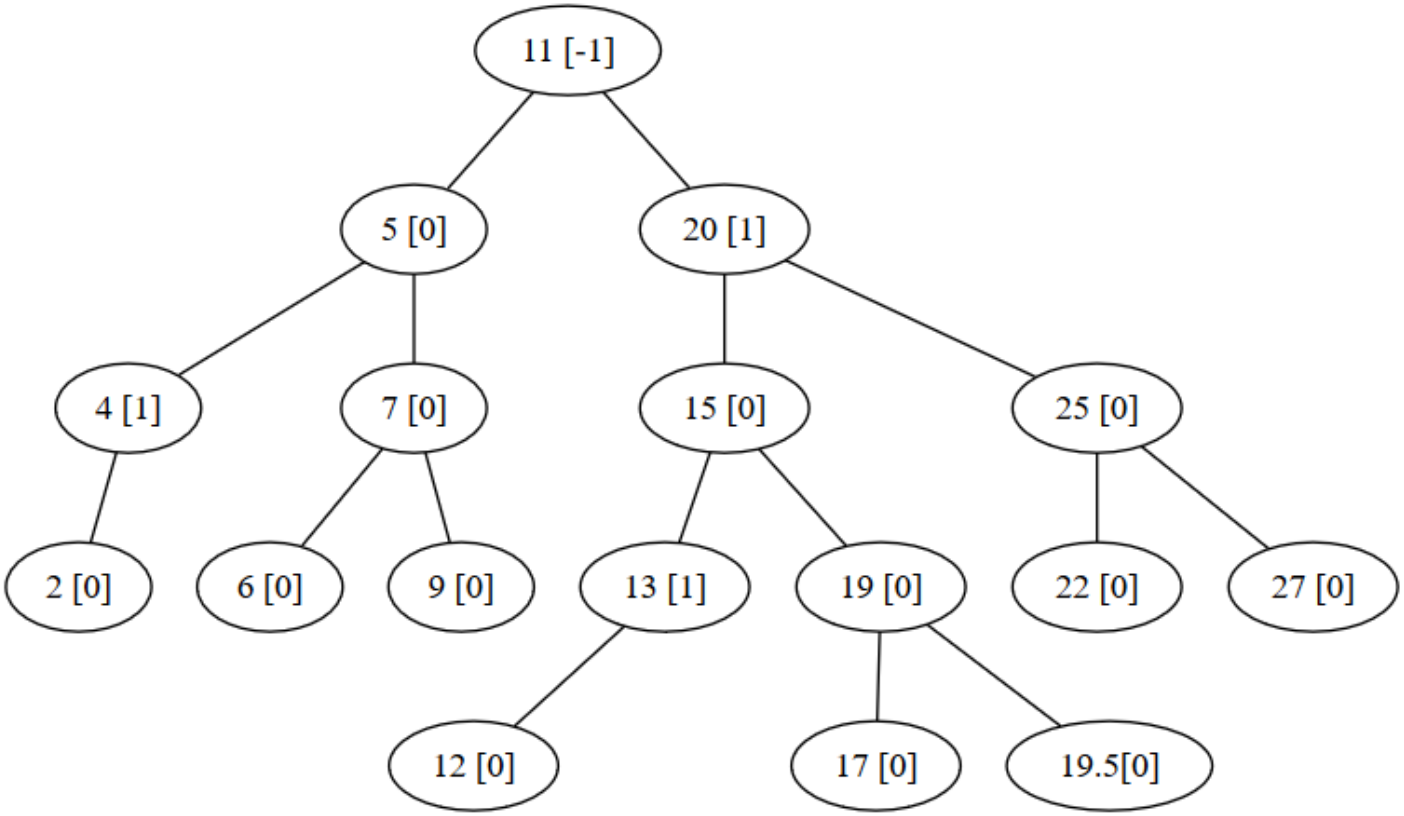
- Insert value 6

AVL Trees - rotations - case 2 example



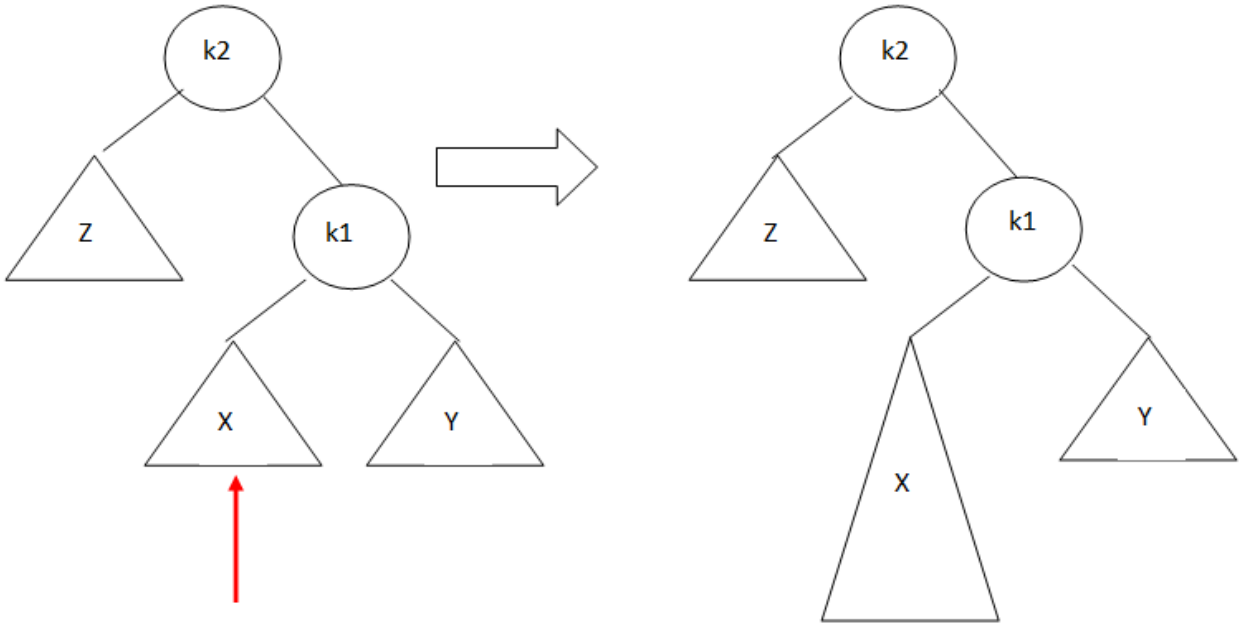
- Node 7 is imbalanced, because we inserted a new node (6) to the right subtree of the left child
- Solution: **double rotation to right**

AVL Trees - rotation - case 2 example



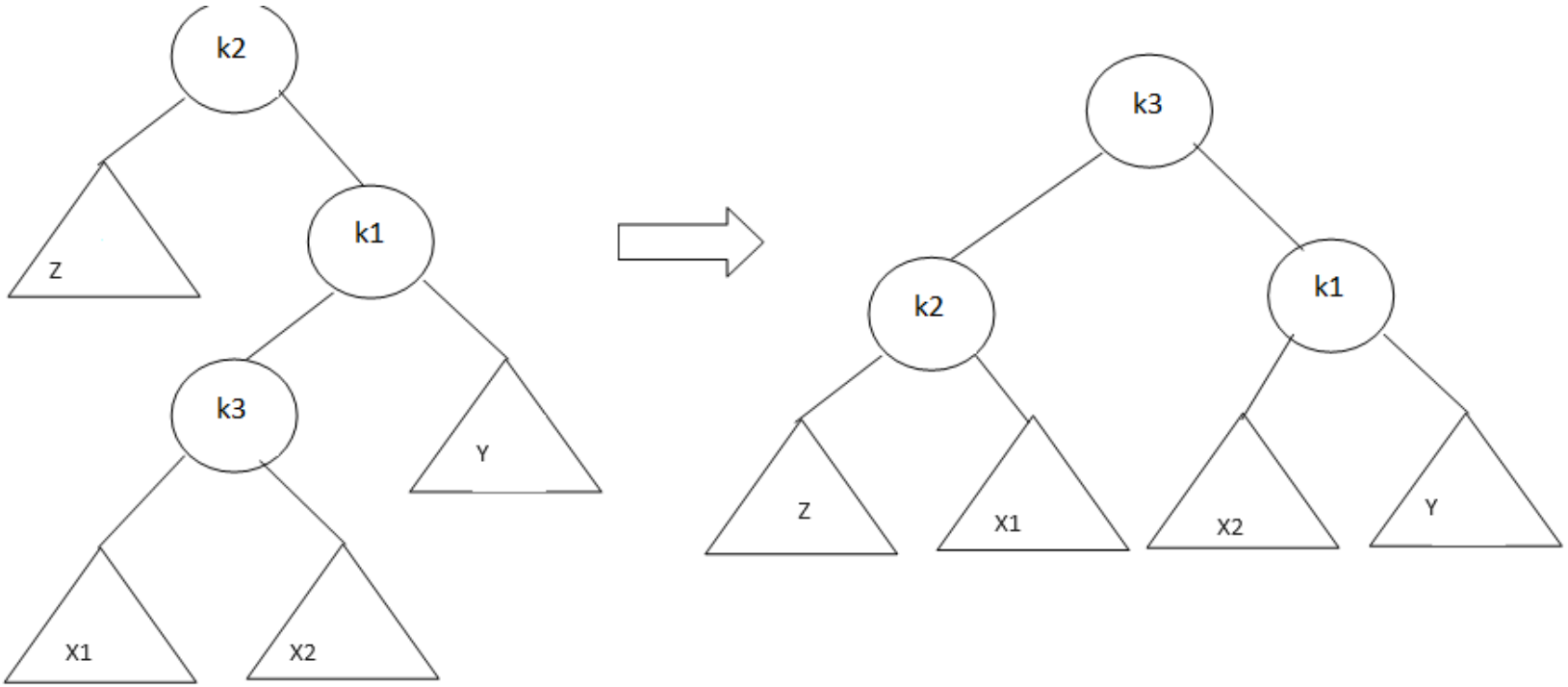
- After the rotation

AVL Trees - rotations - case 3

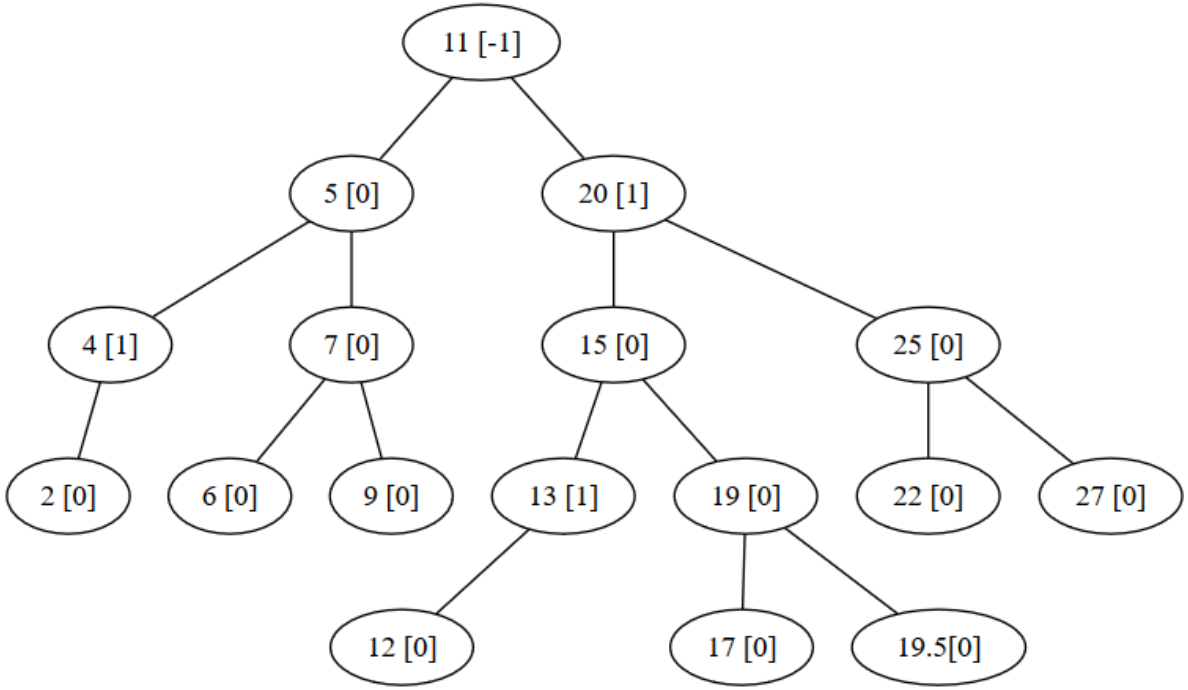


- Solution: **Double rotation to left**

AVL Trees - rotation - Double Rotation to Left

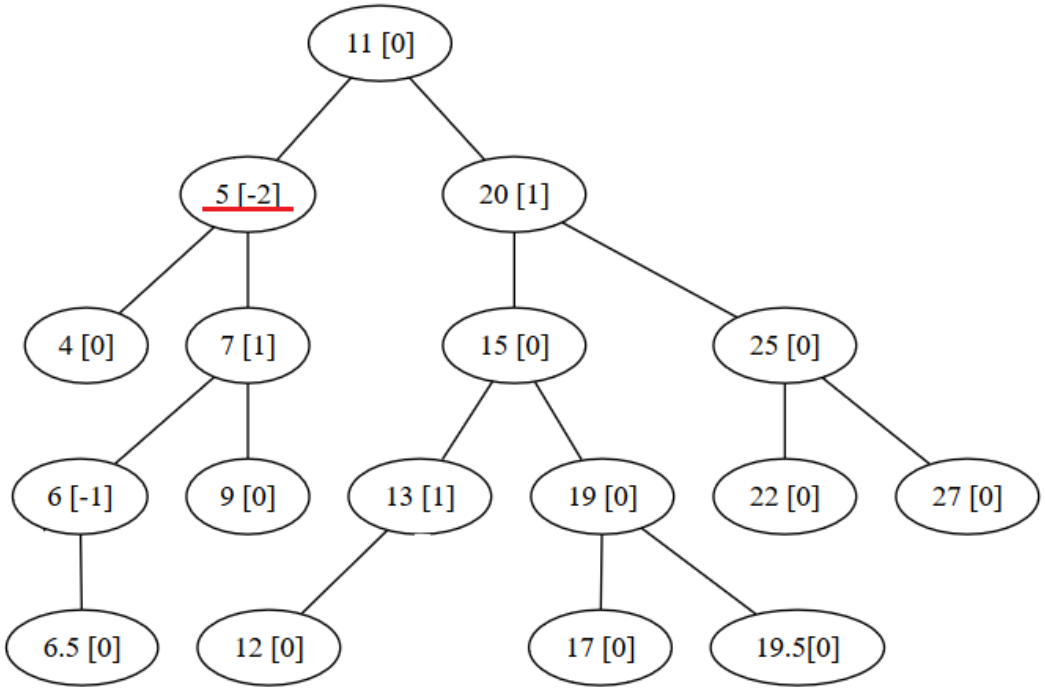


AVL Trees - rotations - case 3 example



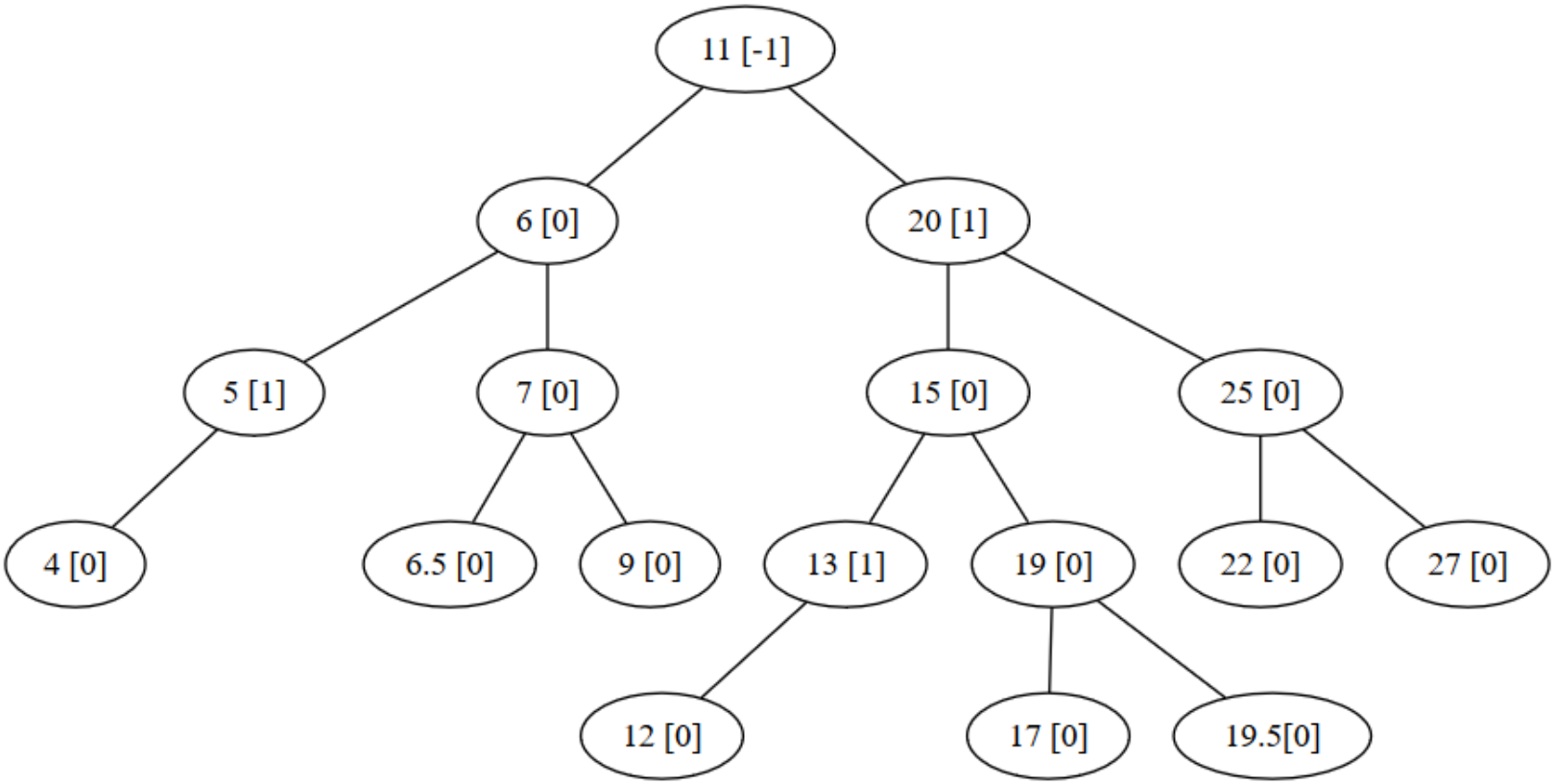
- Remove node with value 2 and insert value 6.5

AVL Trees - rotations - case 3 example



- Node 5 is imbalanced, because we inserted a new node (6.5) to the left subtree of the right child
- Solution: **double rotation to left**

AVL Trees - rotation - case 3 example



- After the rotation

Recognizing the rotations I

- One way of determining what kind of rotation we have is to consider the following terms:
  - *heavy-left* - a node is heavy-left if it has more nodes on the left side than on the right side. $\rightarrow$ a rotation to the right is needed to balance it.
  - *heavy-right* - a node is heavy-right if it has more nodes on the right side than on the left side. $\rightarrow$ a rotation to the left is needed to balance it.

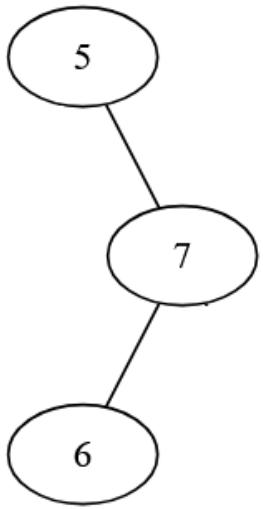
Recognizing the rotations II

- When a node is imbalanced, we first check if it is *heavy-left* or *heavy-right*. If the node is heavy-left we look at its left child to see if it is *heavy-left* (a single notation will be enough) or *heavy-right* (we will need a double rotation).

- We do the same if the node is *heavy-right*, looking at its right child to see if it is *heavy-left* (a double rotation will be needed) or *heavy-right* (a single rotation is enough).

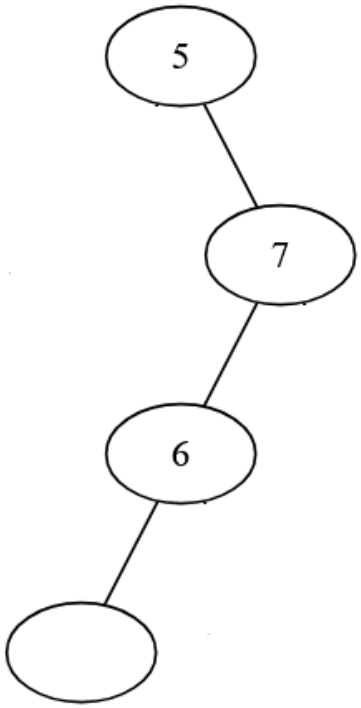
Double rotations I

- Some resources say that a double rotation can be achieved by doing two consecutive single rotations, but for the first one we do not have all the nodes.
- Let's see an example. Consider the following simple case:



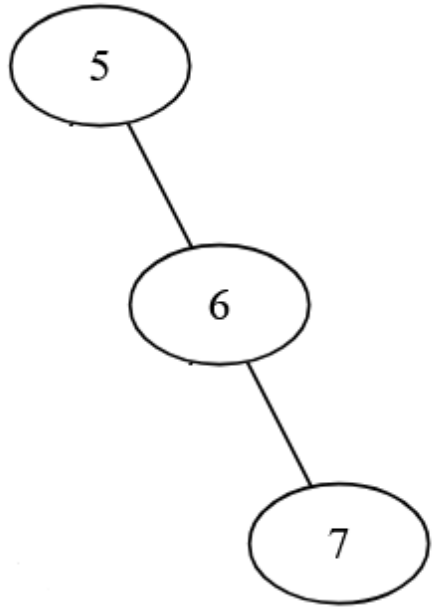
- This is the simplest, classic case of a double left rotation. However, we could also imagine that node 6 has a left child.

Double rotations II



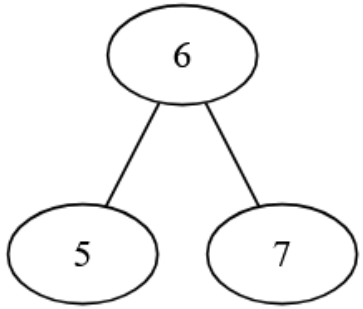
- In this case, we actually have an imbalance at node 7, where we need a single right notation to balance it.

Double rotations III



- Now, we need a single left rotation to balance the tree.

Double rotations IV



- So, we have done a double left rotation by first doing a single right rotation (with incomplete nodes) and then a single left one.

AVL rotations example I

- Start with an empty AVL tree
- Insert 2

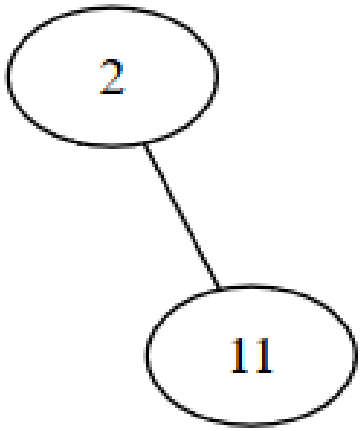
AVL rotations example II

- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example III

- No rotation is needed
- Insert 11

AVL rotations example IV

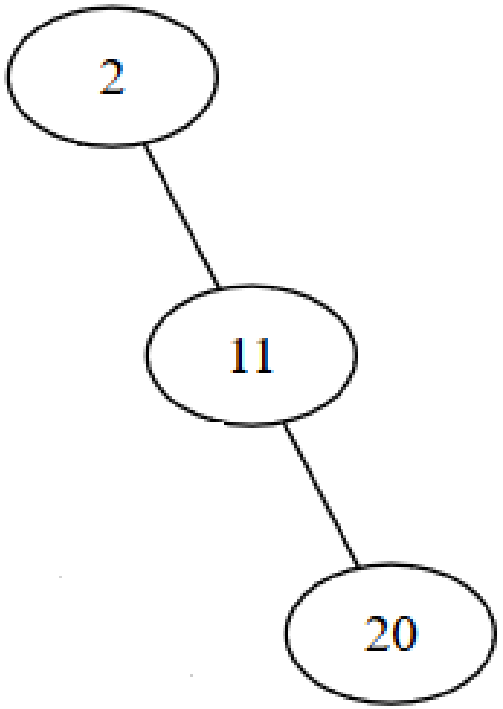


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example V

- No rotation is needed
- Insert 20

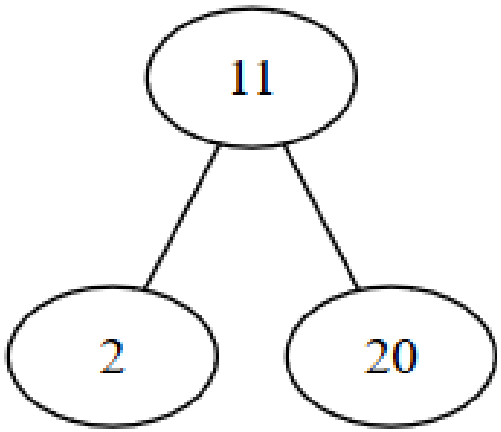
AVL rotations example VI



- Do we need a rotation?
- If yes, on which node and what type of rotation?

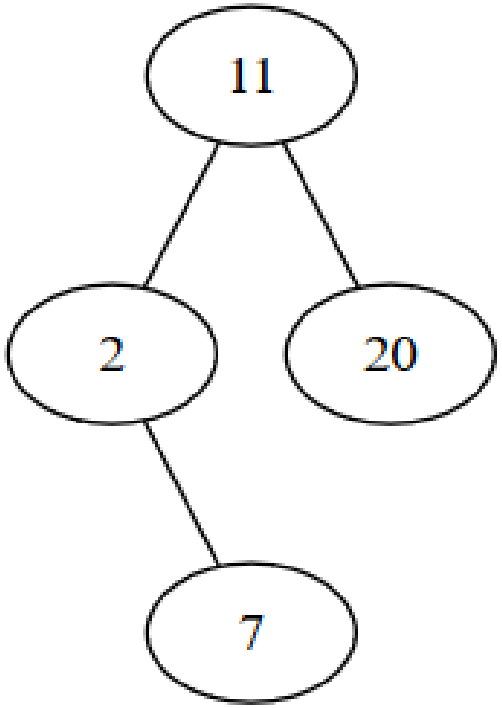
AVL rotations example VII

- Yes, we need a single left rotation on node 2
- After the rotation:



- Insert 7

AVL rotations example VIII

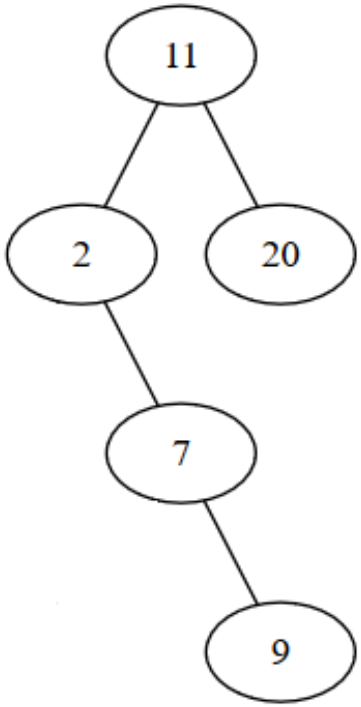


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example IX

- No rotation is needed
- Insert 9

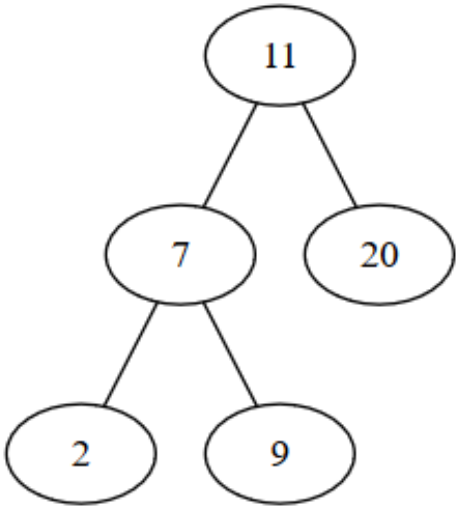
AVL rotations example X



- Do we need a rotation?
- If yes, on which node and what type of rotation?

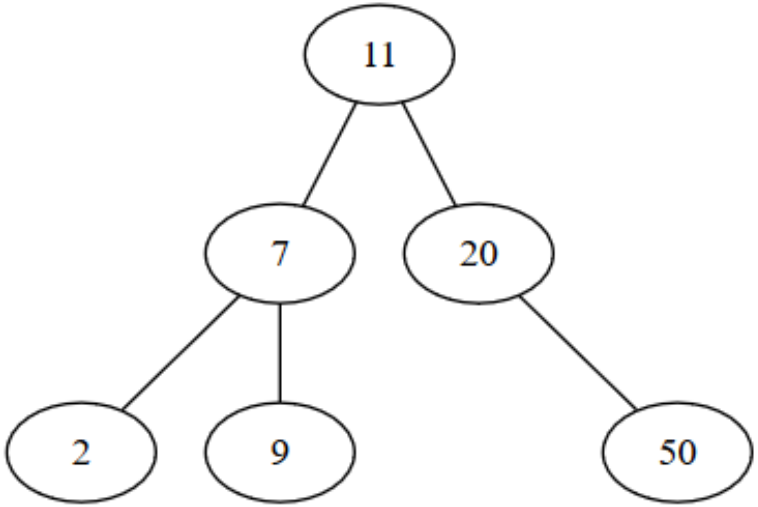
AVL rotations example XI

- Yes, we need a single left rotation on node 2
- After the rotation:



- Insert 50

AVL rotations example XII

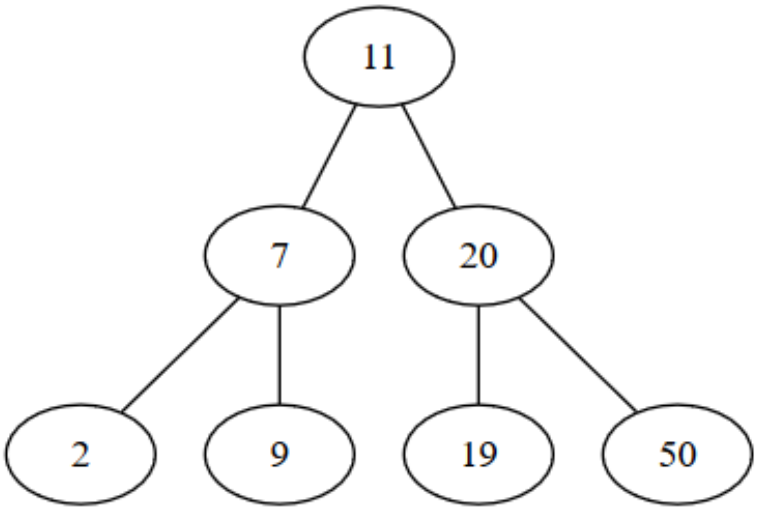


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XIII

- No rotation is needed
- Insert 19

AVL rotations example XIV

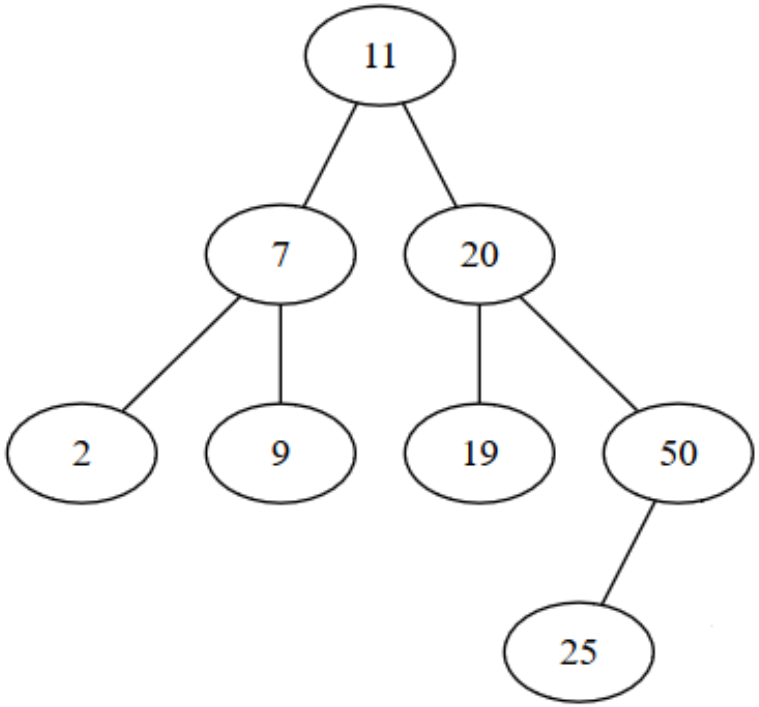


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XV

- No rotation is needed
- Insert 25

AVL rotations example XVI

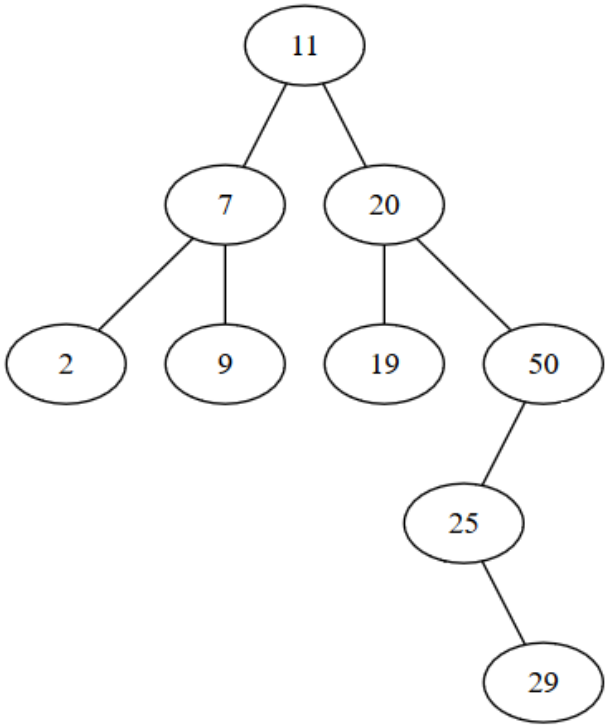


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XVII

- No rotation is needed
- Insert 29

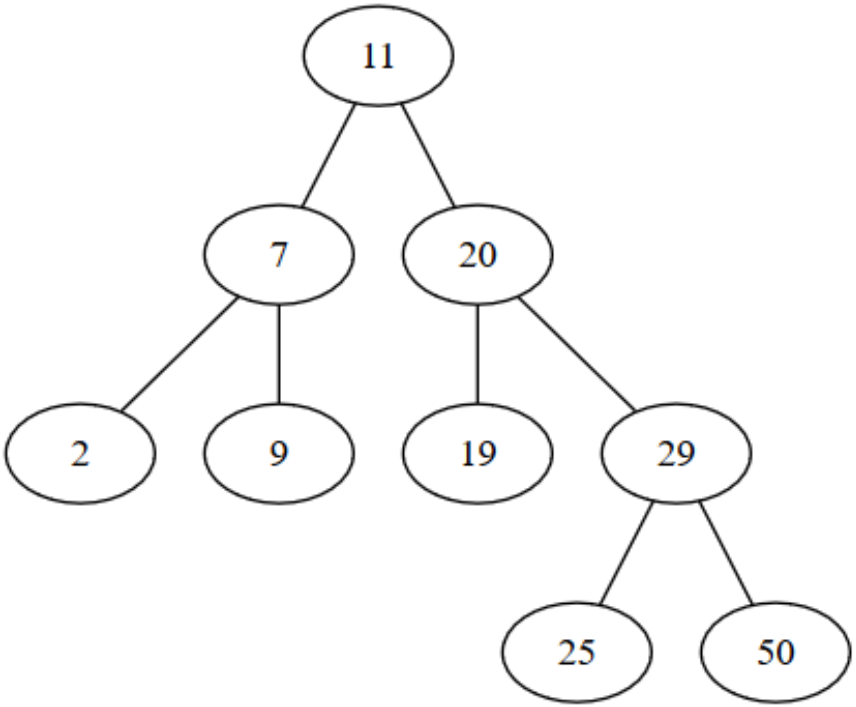
AVL rotations example XVIII



- Do we need a rotation?
- If yes, on which node and what type of rotation?

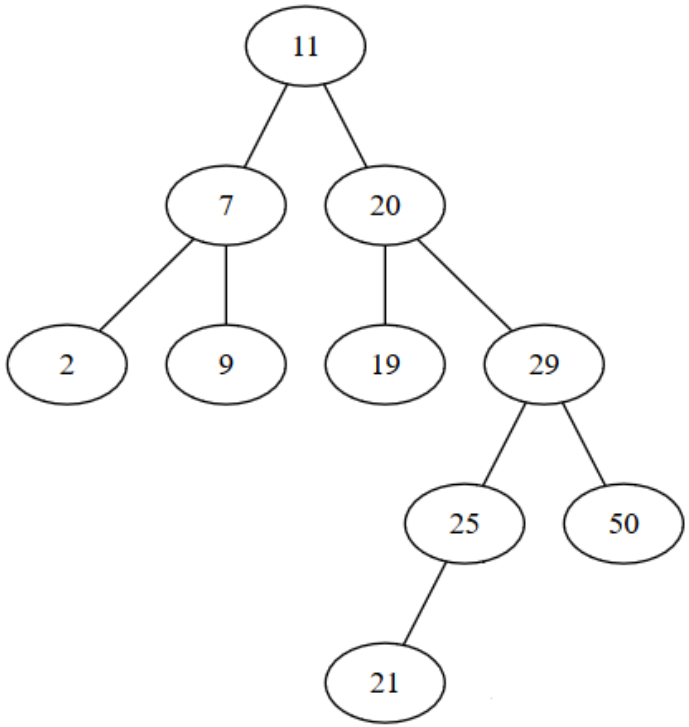
AVL rotations example XIX

- Yes, we need a double right rotation on node 50
- After the rotation



- Insert 21

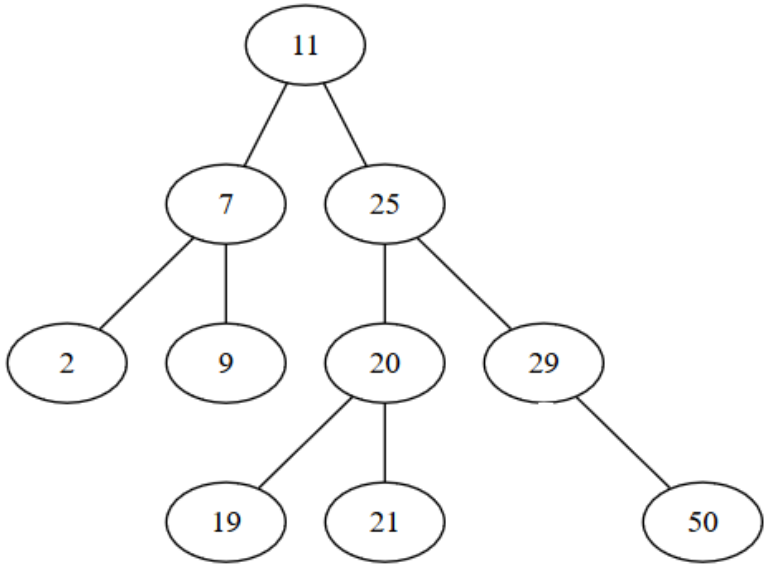
AVL rotations example XX



- Do we need a rotation?
- If yes, on which node and what type of rotation?

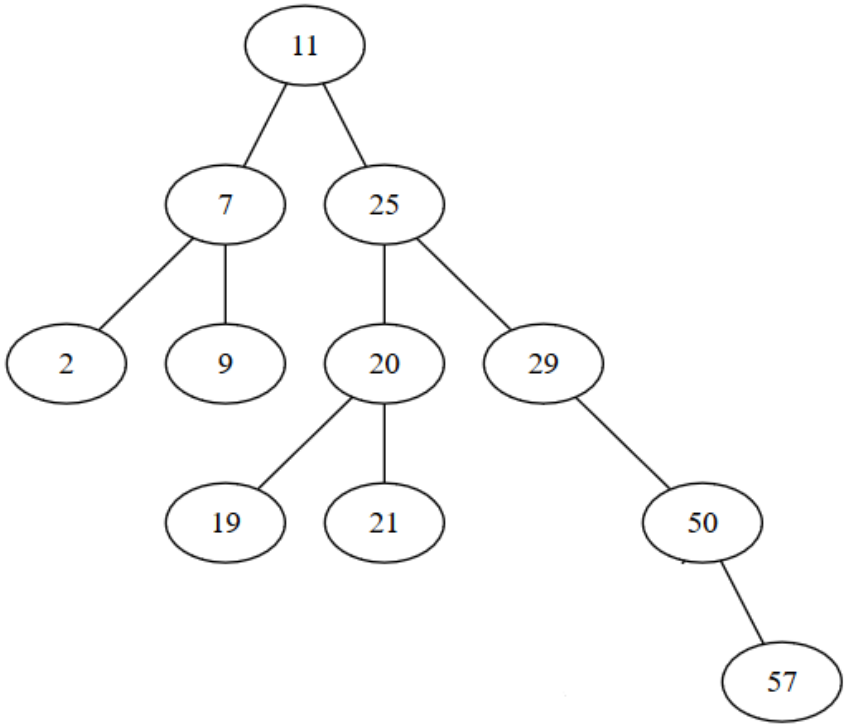
AVL rotations example XXI

- Yes, we need a double left rotation on node 20
- After the rotation



- Insert 57

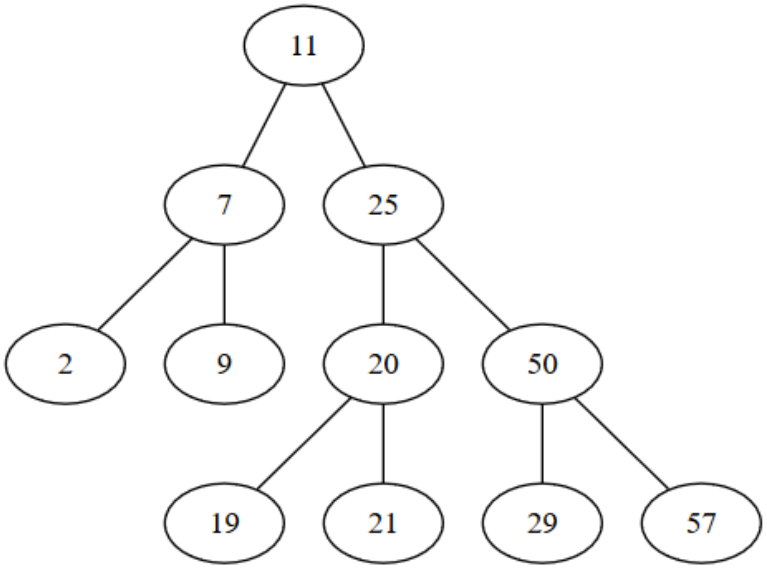
AVL rotations example XXII



- Do we need a rotation?
- If yes, on which node and what type of rotation?

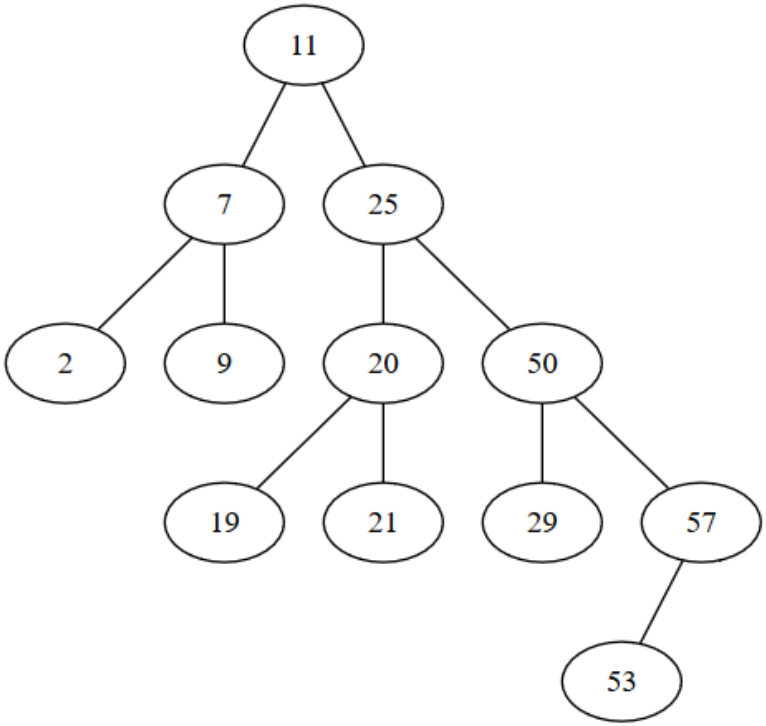
AVL rotations example XXIII

- Yes, we need a single left rotation on node 50
- After the rotation



- Insert 53

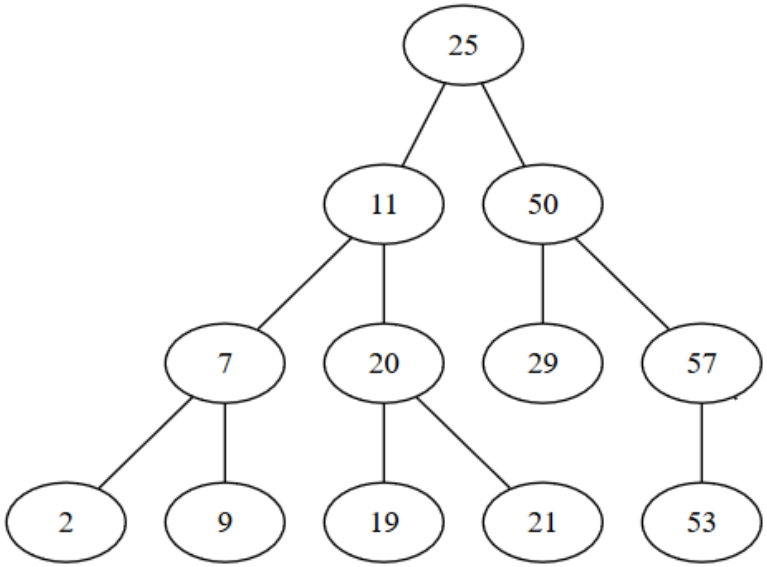
AVL rotations example XXIV



- Do we need a rotation?
- If yes, on which node and what type of rotation?

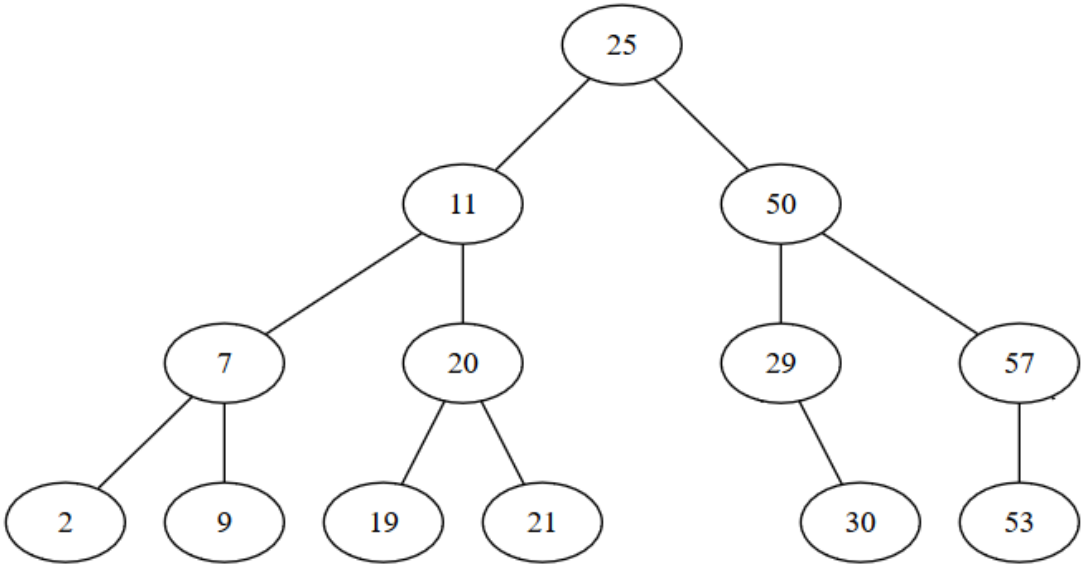
AVL rotations example XXV

- Yes, we need a single left rotation on node 11
- After the rotation



- Insert 30

AVL rotations example XXVI

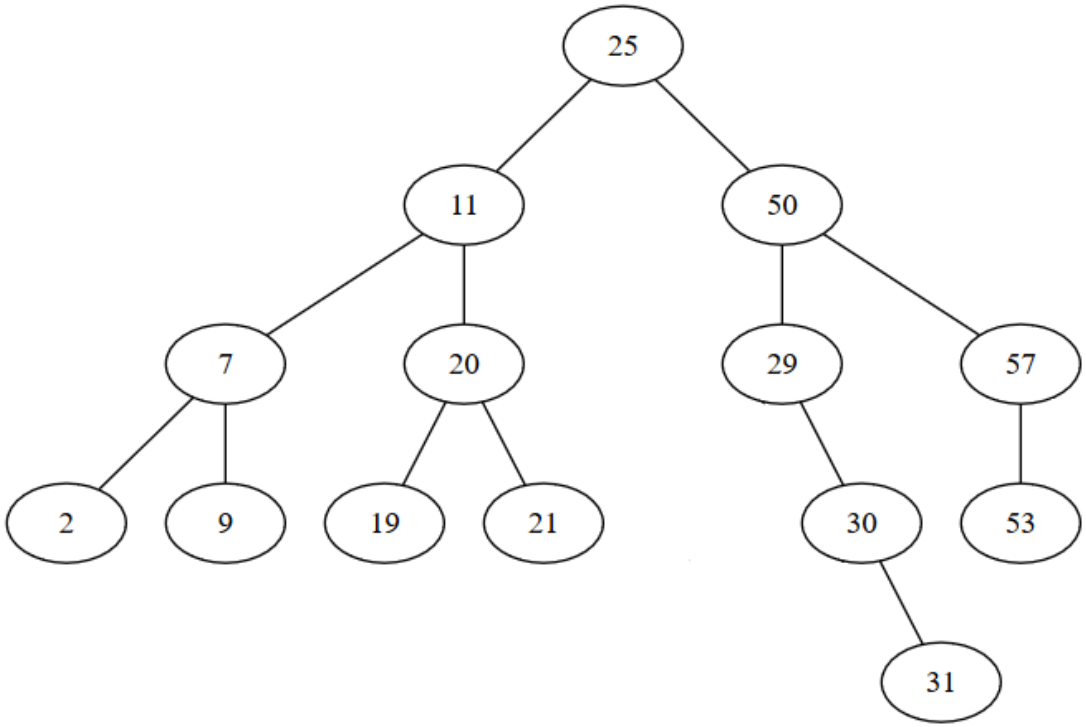


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XXVII

- No rotation is needed
- Insert 31

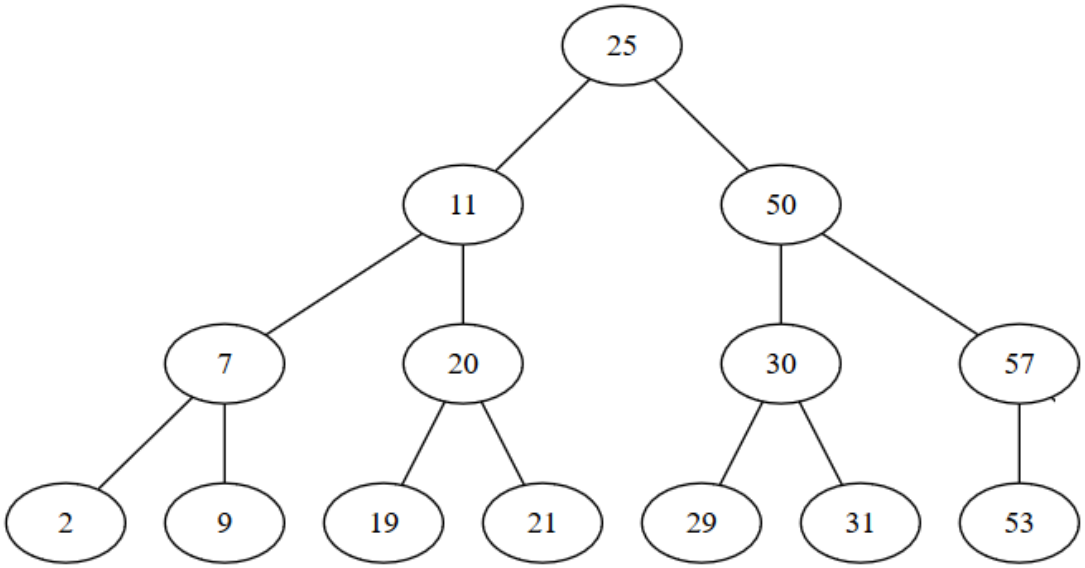
AVL rotations example XXVIII



- Do we need a rotation?
- If yes, on which node and what type of rotation?

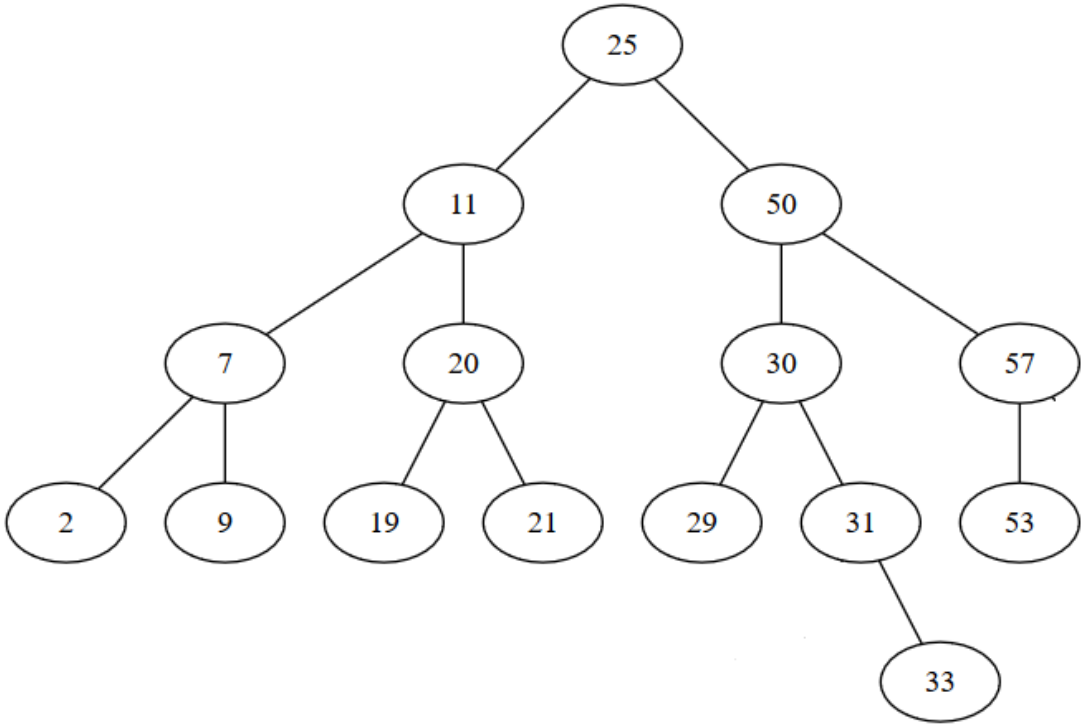
AVL rotations example XXIX

- Yes, we need a single left rotation on node 29
- After the rotation



- Insert 33

AVL rotations example XXX

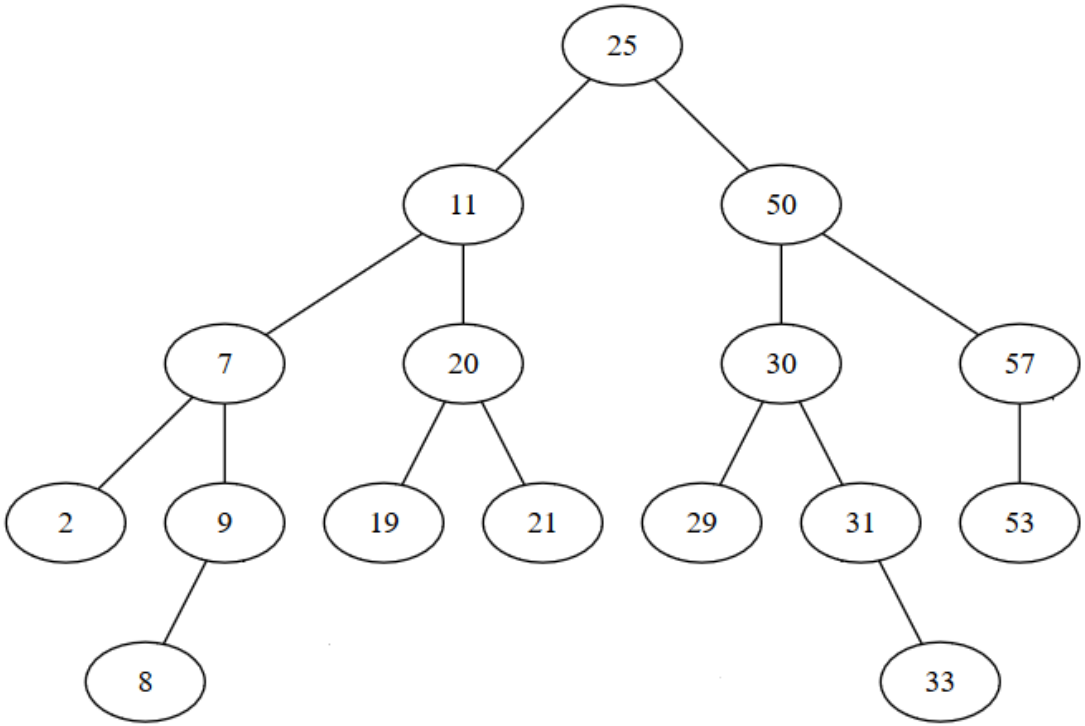


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XXXI

- No rotation is needed
- Insert 8

AVL rotations example XXXII

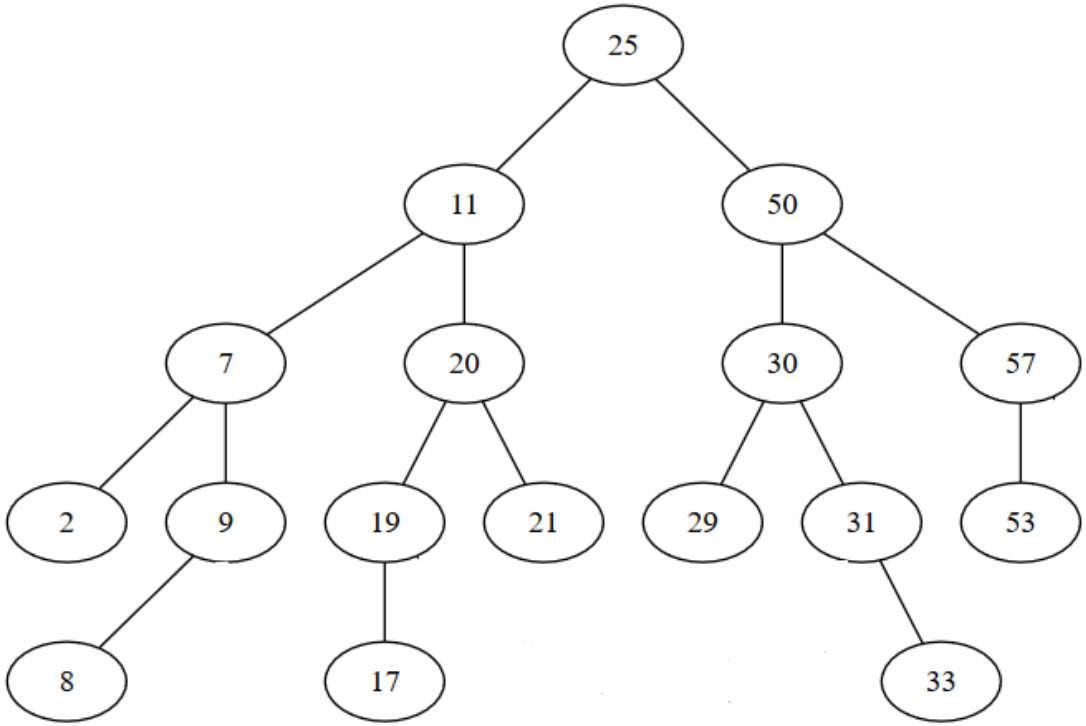


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XXXIII

- No rotation is needed
- Insert 17

AVL rotations example XXXIV

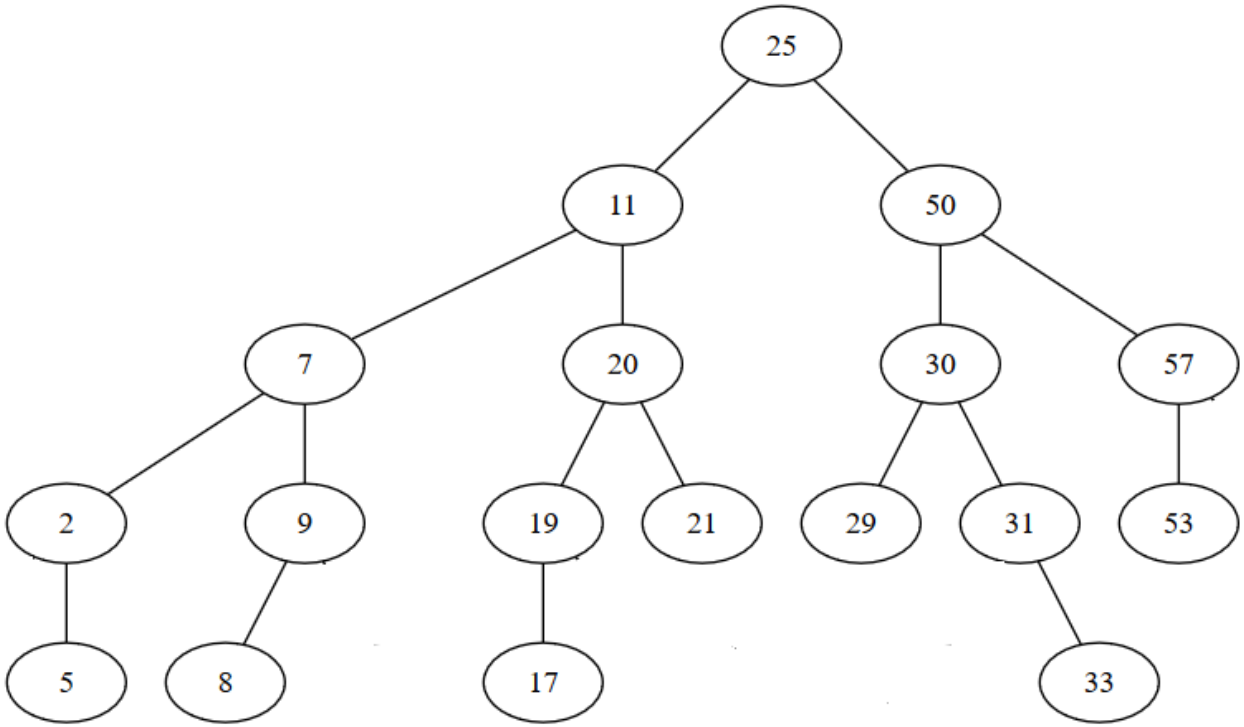


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XXXV

- No rotation is needed
- Insert 5

AVL rotations example XXXVI

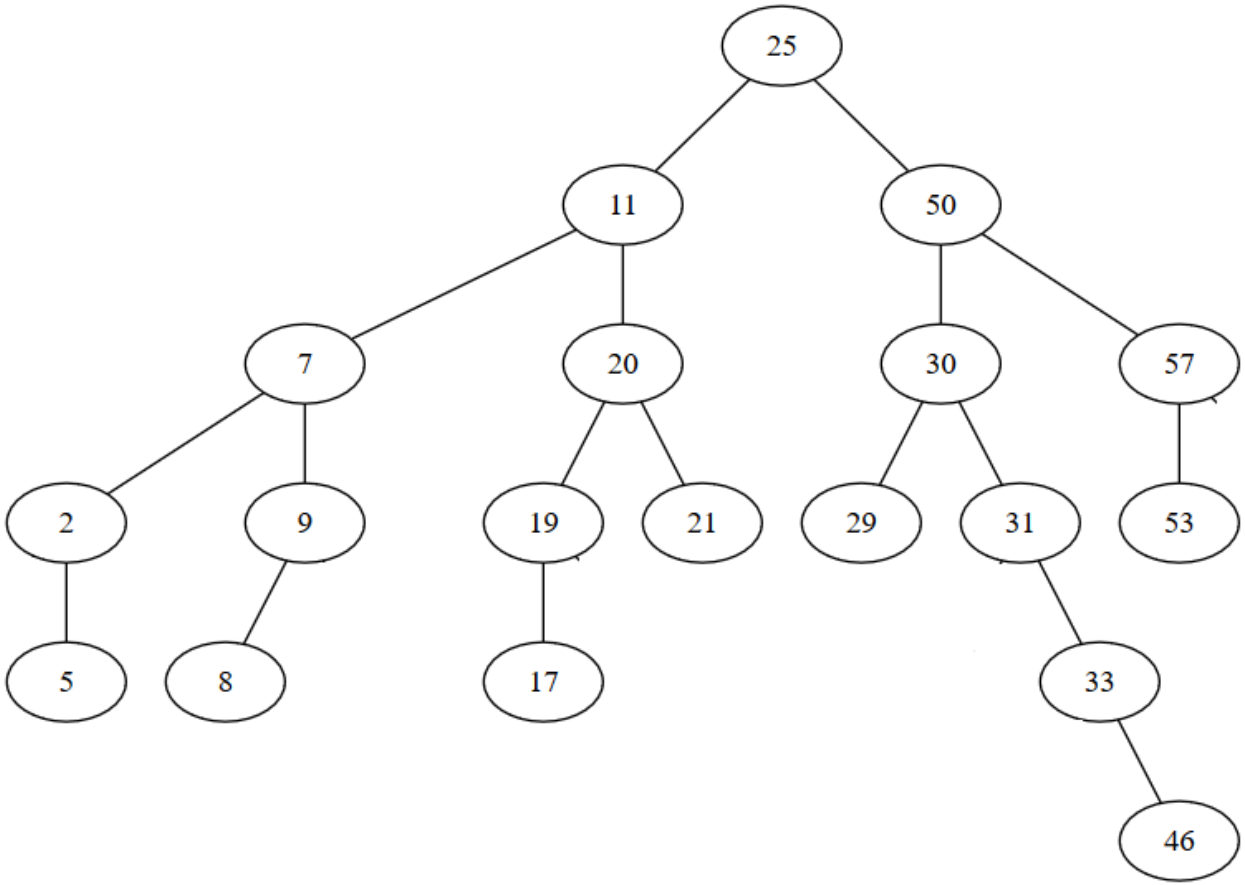


- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XXXVII

- No rotation is needed
- Insert 46

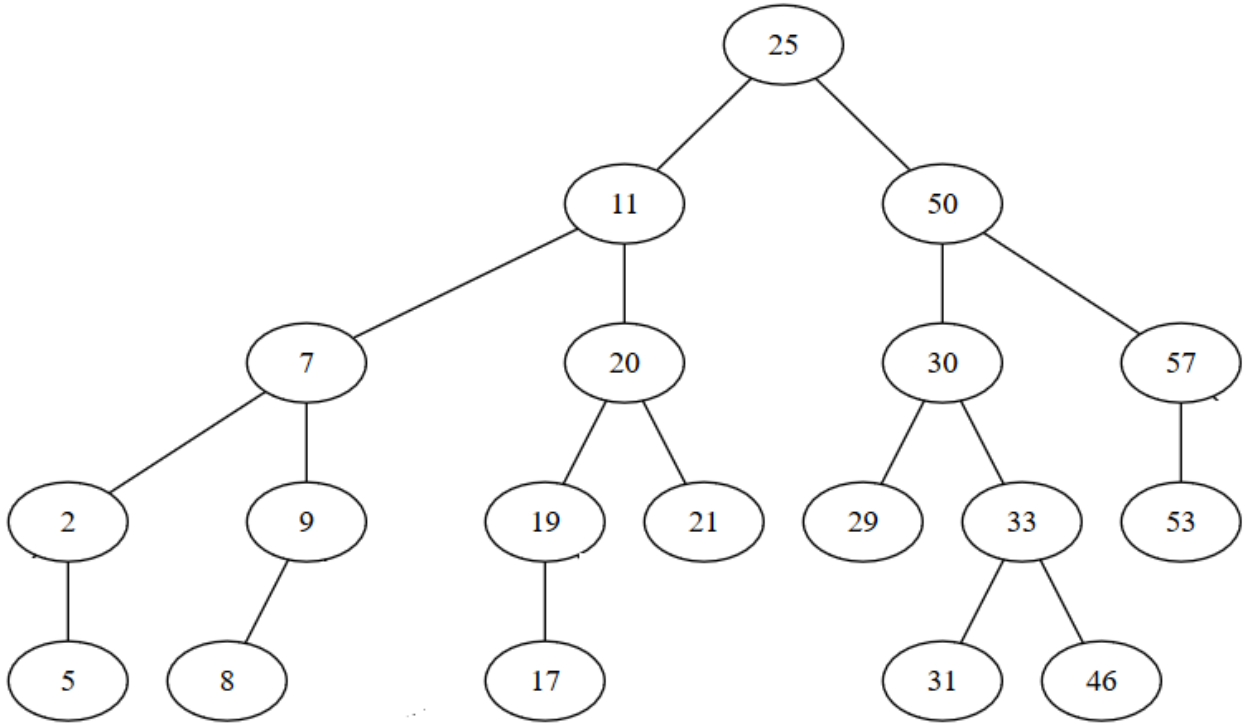
AVL rotations example XXXVIII



- Do we need a rotation?
- If yes, on which node and what type of rotation?

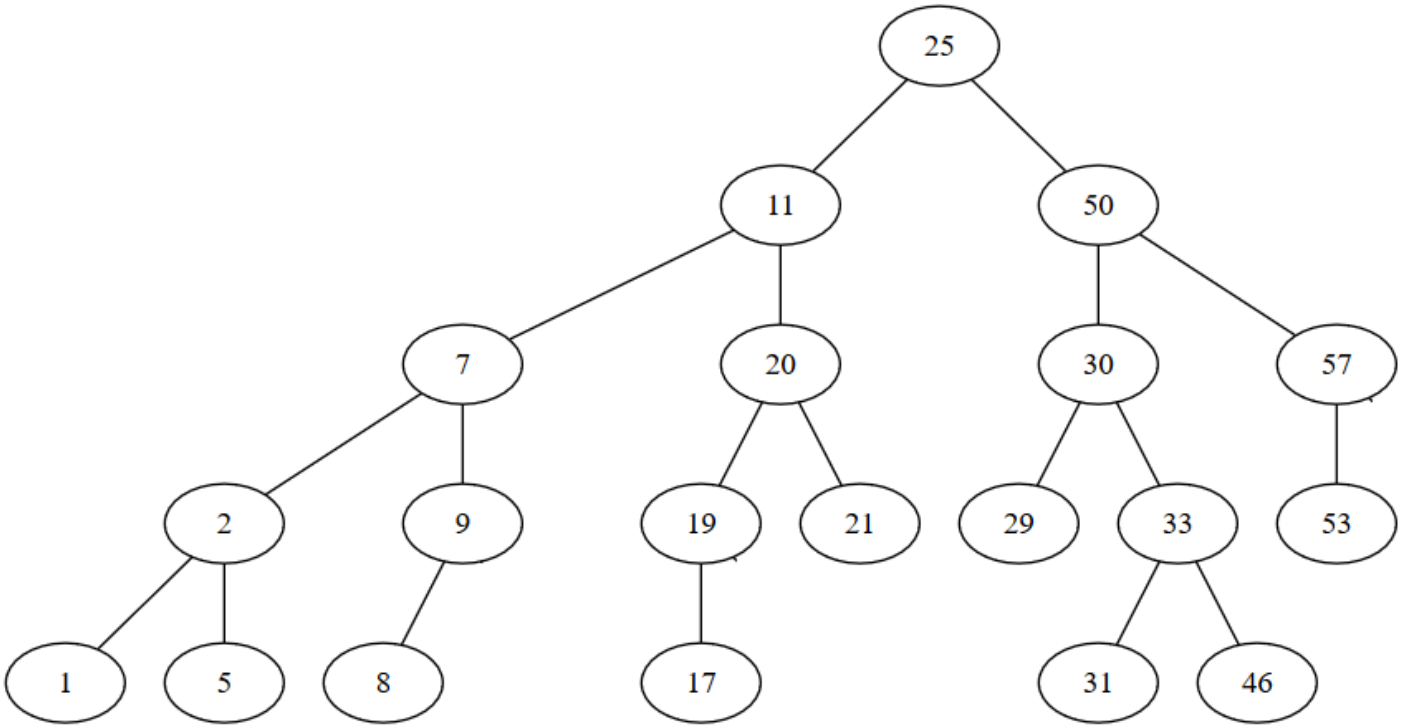
AVL rotations example XXXIX

- Yes, we need a single left rotation on node 31
- After the rotation



- Insert 1

AVL rotations example XL



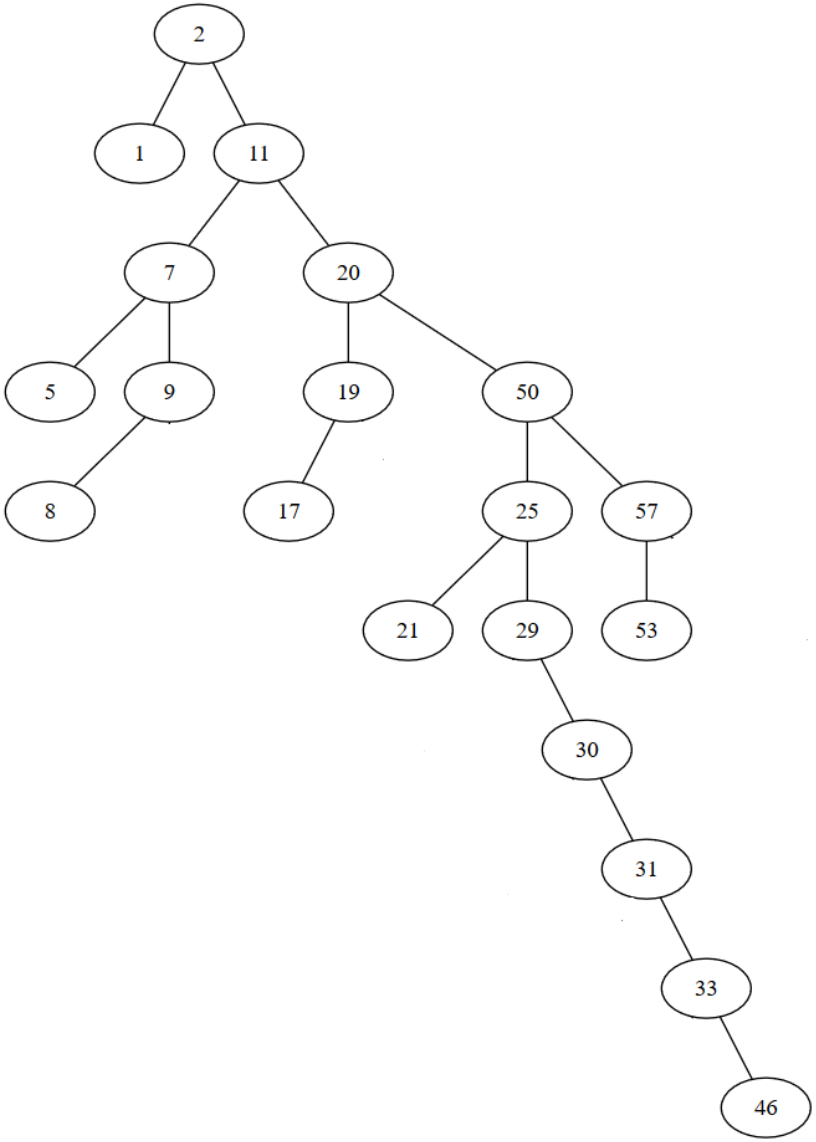
- Do we need a rotation?
- If yes, on which node and what type of rotation?

AVL rotations example XLI

- No rotation is needed

Comparison to BST

- If, instead of using an AVL tree, we used a binary search tree, after the insertions the tree would have been:


2
1
11
7
20
5
9
19
50
8
17
25
57
21
29
53
30
31
33
46

Summary

- Today we have talked about:
  - Binary Trees
  - AVL Trees